

DISCELL: Supplementary Material

Guide

Sections follow the main text: Secs. S1 and S2 back Sec. 2, Secs. S3 and S4 Sec. 3.1, Sec. S5 Sec. 3.2, Sec. S6 Secs. 3.3 to 3.6, Sec. S7 Sec. 3.7, and Sec. S8 collects the limitations.

Claim	Main text	Check in the supplement
Model, objective, bound on κ	Sec. 2	Sec. S1, proof Sec. S1.3, symmetries Sec. S1.4; settings Sec. S2.2
C1 Spill-over is bounded, not estimated; breakdown points	Secs. 2.4 and 3.7, Table 4	Sec. S7.1: Table S20, Fig. S6, Table S21; Sec. S7.3; Sec. S7.4
C2 Less residual niche signal, more intrinsic state	Sec. 3.5, Fig. 4	Table S9, Table S10; methods Sec. S3.3
C3 $\mathbf{z}$ keeps type and cycle	Sec. 3.3	Table S12, Table S9; definitions Sec. S4.1
C4 $\mathbf{w}$ carries the niche; sublabels split	Sec. 3.3	Sec. S6.1, Sec. S6.2, Sec. S4.2; limits Sec. S5.2, Sec. S7.5
C4b Response programmes	Sec. 3.4	Fig. S4, Sec. S6.1
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S1 Model: Notation, Derivations and Proofs

Table S1: Notation.

Symbol	Meaning
$\mathbf{x}_i, \ell_i$	count vector of cell i over G genes; its total, conditioned on and never modelled; $\bar{\ell}$ the section's median total
$t_i \in \{1, \dots, K\}$	lineage label
$\mathbf{z}_i \in \mathbb{R}^{d_z}$	intrinsic state; prior $\mathcal{N}(\mathbf{0}, \mathbf{I})$
$\mathbf{w}_i \in \mathbb{R}^{d_w}$	response; prior $\mathcal{N}(m_\psi(\mathbf{c}_i, t_i), \sigma_w^2 \mathbf{I})$, $\sigma_w = 1$
$\mathbf{c}_i$	niche descriptor: attention over the neighbours' types $\oplus$ masked image embedding $\Phi_i \oplus$ isolation flag
$\mathbf{y}_i, \Phi_i$	neighbour-type composition; image embedding with a disc over the cell set to zero
ρ_i	clean composition softmax($a(\mathbf{z}_i) + \mathbf{B}\mathbf{w}_i$); a the decoder, $\mathbf{B} \in \mathbb{R}^{G \times d_w}$ the loadings
$\mathcal{N}_i, \beta_{ij}$	neighbour set in the pruned Delaunay graph; fixed spill-over kernel
$\bar{\rho}_i$	spill-over influx $\sum_{j \in \mathcal{N}_i} \beta_{ij} \rho_j$
$\kappa, \mathbf{p}_i$	spill-over fraction, fixed per fit and swept; fitted composition $(1 - \kappa)\rho_i + \kappa\bar{\rho}_i$
$\bar{\kappa}_i$	largest spill-over fraction compatible with cell i 's composition (Proposition 1)
$\mu_{z,i}, \mu_{w,i}$	posterior means, which the probes and readouts use
$\alpha_z, \alpha_w, \alpha_a, \omega$	weights of the $\mathbf{z}$ -divergence, the $\mathbf{w}$ -divergence, the adversary and the intrinsic path

S1.1 The Reconstruction Bound

Fix a cell i and hold $\mathbf{c}_i$ and $\bar{\boldsymbol{\rho}}_i$ constant, as Sec. 2.2 does (we suppress the conditioning on $\bar{\boldsymbol{\rho}}_i$). Under the variational family $q(\mathbf{z})q(\mathbf{w} \mid \mathbf{c}_i, t_i, \mathbf{z}, \mathbf{x}_i)$, Jensen’s inequality gives

$$\log p(\mathbf{x}_i \mid \mathbf{c}_i, t_i) \geq \mathbb{E}_{q(\mathbf{z})} \left[\mathbb{E}_{q(\mathbf{w} \mid \mathbf{z})} [\log p(\mathbf{x}_i \mid \mathbf{z}, \mathbf{w})] - \text{KL}(q(\mathbf{w} \mid \mathbf{z}) \parallel p(\mathbf{w} \mid \mathbf{c}_i, t_i)) \right] - \text{KL}(q(\mathbf{z}) \parallel p(\mathbf{z})) =: L_i. \quad (\text{S1})$$

Because $q(\mathbf{w} \mid \cdot)$ conditions on the sampled $\mathbf{z}$, the $\mathbf{w}$ -divergence sits inside $\mathbb{E}_{q(\mathbf{z})}$; evaluating the closed-form Gaussian divergence at the single reparameterised draw of $\mathbf{z}$ is an unbiased one-sample estimate. Summed over cells, with each neighbour’s latents drawn from its own posterior (Sec. S2.5), $\sum_i L_i$ estimates the evidence lower bound of the coupled model under a fully factorised family. The per-cell bounds are not independent, since $\bar{\boldsymbol{\rho}}_i$ depends on the neighbours’ latents; the stop-gradient removes the cross terms from the gradient, so training is a fixed-point iteration in which each step treats the current $\bar{\boldsymbol{\rho}}$ as data.

Isolated cells. For a cell with no neighbours after pruning, $\bar{\boldsymbol{\rho}}_i = \mathbf{0}$ and the mixture of Eq. (5) would sum to $1 - \kappa$, charging the cell for spill-over it cannot receive. The mixture is therefore renormalised per row,

$$\mathbf{p}_i = \frac{(1 - \kappa) \boldsymbol{\rho}_i + \kappa \bar{\boldsymbol{\rho}}_i}{(1 - \kappa) + \kappa \sum_{j \in \mathcal{N}_i} \beta_{ij}}, \quad (\text{S2})$$

so that an isolated cell has $\mathbf{p}_i = \boldsymbol{\rho}_i$; for connected cells the denominator is one. Probabilities are floored at 10^{-8} inside the logarithm. The graph part of an isolated cell’s niche descriptor is set to zero and its isolation flag raised; the image part stays. Isolated cells are left out of the type-mean targets $\bar{\mathbf{y}}(t)$ and $\bar{\boldsymbol{\Phi}}(t)$ but not out of the adversary or the probes.

Why a multinomial. Independent Poisson counts, conditioned on their total ℓ_i , are multinomial; since ℓ_i is observed, the multinomial needs no library-size latent and does not count depth twice. The mixture of Eq. (5) is exact if the cell’s own and the spill-over transcripts are independent Poisson sources and the spill-over rate scales with the receiving cell’s own rate (Sec. S8). Negative-binomial sources are not closed under addition, and a dispersion on the mixed counts would absorb part of what the mixture should explain. At a median of 178 to 1,401 counts per cell over five thousand genes (Table S6), gene-wise overdispersion cannot be estimated in any case; the natural extension is a Dirichlet-multinomial with one concentration scalar.

S1.2 The Intrinsic Path and the Status of the Objective

Restricting the variational family to $q(\mathbf{w}) := p(\mathbf{w} \mid \mathbf{c}_i, t_i)$ gives a second bound on the same evidence,

$$\log p(\mathbf{x}_i \mid \mathbf{c}_i, t_i) \geq \mathbb{E}_{q(\mathbf{z})} \mathbb{E}_{p(\mathbf{w} \mid \mathbf{c}_i, t_i)} [\log p(\mathbf{x}_i \mid \mathbf{z}, \mathbf{w})] - \text{KL}(q(\mathbf{z}) \parallel p(\mathbf{z})) =: L_i^z.$$

Each bound carries its own $\mathbf{z}$ -divergence, so $L_i + \omega L_i^z \leq (1 + \omega) \log p(\mathbf{x}_i \mid \mathbf{c}_i, t_i)$ contains $-(1 + \omega) \text{KL}(q(\mathbf{z}_i) \parallel p(\mathbf{z}_i))$, with gap $\text{KL}(q(\mathbf{z})q(\mathbf{w} \mid \mathbf{z}) \parallel p(\mathbf{z}, \mathbf{w} \mid \cdot)) + \omega \text{KL}(q(\mathbf{z})p(\mathbf{w} \mid \mathbf{c}_i, t_i) \parallel p(\mathbf{z}, \mathbf{w} \mid \cdot))$. This mirrors the two-bound objective of Slavutsky et al. [1], except that both of our terms bound the same evidence. One $q(\mathbf{z})$ serves two families, so its optimum is a compromise. The path is the only term that rewards $\mathbf{z}$ for being informative, since the prior on $\mathbf{w}$ never references $\mathbf{z}$; it decodes through the same spill-over mixture at the same κ . At the final configuration it acts on the allocation: without it, $\mathbf{w}$ takes up type signal (Sec. S7.4).

Scaling of the weights. The reconstruction terms of Eq. (8) are divided by ℓ_i , so that weights transfer between sections of different depth; reconstruction becomes $O(1)$ while the divergences stay absolute. At $\alpha_z = \alpha_w = 1/\bar{\ell}$ and $\alpha_a = 0$, J is proportional to $L_i + \omega L_i^z$ for a cell of median depth, and that point is the reference for the weights; $\alpha_z = \frac{1}{2}/\bar{\ell}$ restores a single charge of the $\mathbf{z}$ -divergence at $\omega = 1$. The response weight departs from it deliberately ($\alpha_w = 0.1$; Sec. S2.3). At that weight the response posterior stays close to its prior mean, so the response readouts are in effect readouts of the context regression $m_\psi(\mathbf{c}_i, t_i)$, with exceptions in subtype recovery (Sec. S4.2); a planted per-cell response is not detected at this weight (Sec. S5.2). The per-cell scaling also weights a cell’s divergences, relative to its own bound, in proportion to its depth (Sec. S6.3), the price of transferable weights.

Status of the objective. Once a weight departs from $1/\bar{\ell}$, J is a weighted surrogate rather than a bound. Neighbour compositions enter under a stop-gradient and, with the adversary, the fit is a saddle point of a two-player game, so training does not maximise J as a single function. The invariance term is part of no bound: it is posterior regularisation [2], a constraint $I(\mathbf{z}; [\mathbf{y}, \Phi] \mid t) = 0$ on the aggregate of the encoder’s outputs, the status Slavutsky et al. [1] give their adversarial term; the generative model does not model $\mathbf{y}$ or Φ . We therefore claim no likelihood-based uncertainty for q : the reported variability comes from the spill-over sweep and the seeds. κ is not a hyperparameter but the sweep axis.

S1.3 The Spill-over Fraction: Proof of Proposition 1 and the Grid

Proof of Proposition 1. The clean composition implied by κ' is $\rho_i(\kappa') = (\mathbf{p}_i - \kappa' \bar{\rho}_i)/(1 - \kappa')$, which sums to one for every κ' and is non-negative in gene g if and only if $\kappa' \bar{\rho}_{ig} \leq p_{ig}$; this holds for every gene if and only if $\kappa' \leq \bar{\kappa}_i$. Since $p_{ig}/\bar{\rho}_{ig} = \kappa + (1 - \kappa)\rho_{ig}/\bar{\rho}_{ig} \geq \kappa$, the true value lies below the bound, and meets it exactly when a gene the neighbours express has zero share in the cell’s own composition. $\square$

A section-wide κ must respect the bound of every cell, and the data pin it only through a gene the cell cannot express itself or through a second view of the cell. With counts the bound is soft, since a κ above it charges a cell for neighbour transcripts it does not show. The bound holds with the neighbours’ clean profiles fixed and is not computed on our sections, so a contrast that survives the grid is shown to survive spill-over fractions up to 0.4, not every fraction the data allow. The grid runs from zero, the nested no-correction null, to 0.4: a typical cell on this platform carries misassigned transcripts of about a tenth of its counts [16], which the operating point 0.1 matches and the top of the grid exceeds several times. Estimating κ would need two views of one cell at different spill-over levels, such as a nuclear/extranuclear split of the transcripts, with a multinomial over $2G$ bins; we leave it as future work.

S1.4 Symmetries of the Response

Given κ , the response coordinates carry at least the symmetries of Proposition 1; we do not show that there are no others. The statement treats a , m_ψ and the posterior networks as arbitrary functions of their inputs, each map is applied to every cell at once, and the response posterior is $\mathcal{N}(\mu_{w,i}, \text{diag } \nu_{w,i}^2)$.

Proposition 1 (Symmetries of the response). *Fix κ , the graph and the kernel.*

- (a) Rotation. *For orthogonal $\mathbf{Q} \in \mathbb{R}^{d_w \times d_w}$, the map $\mathbf{w}_i \mapsto \mathbf{Q}\mathbf{w}_i$, $\mathbf{B} \mapsto \mathbf{B}\mathbf{Q}^\top$, $m_\psi \mapsto \mathbf{Q}m_\psi$ leaves the distribution of the counts in Eqs. (1) to (6) unchanged, and, with each posterior mapped to $\mathcal{N}(\mathbf{Q}\mu_{w,i}, \mathbf{Q}\text{diag } \nu_{w,i}^2 \mathbf{Q}^\top)$, every term of the objective J of Eq. (8). The mapped posterior is again diagonal, so J is unchanged within the diagonal family, when $\mathbf{Q}$ is a signed permutation.*
- (b) Constants over genes. *For any scalar function $c(\mathbf{z})$ and any $\mathbf{b} \in \mathbb{R}^{d_w}$, the map $a \mapsto a + c \mathbf{1}$, $\mathbf{B} \mapsto \mathbf{B} + \mathbf{1}\mathbf{b}^\top$ leaves every clean composition, and hence J , unchanged.*
- (c) Per-type translation. *Suppose there is a function τ with $\tau(\mathbf{z}) = t_i$ for every $\mathbf{z}$ in the support of $q(\mathbf{z}_i)$, for every cell i . For any $\delta_1, \dots, \delta_K \in \mathbb{R}^{d_w}$, the map $\mathbf{w}_i \mapsto \mathbf{w}_i + \delta_{t_i}$ (the posterior mean with it), $m_\psi(\mathbf{c}, t) \mapsto m_\psi(\mathbf{c}, t) + \delta_t$, $a(\mathbf{z}) \mapsto a(\mathbf{z}) - \mathbf{B}\delta_{\tau(\mathbf{z})}$ leaves J unchanged.*

A rescaling $\mathbf{w} \mapsto \mathbf{S}\mathbf{w}$, $\mathbf{B} \mapsto \mathbf{B}\mathbf{S}^{-1}$ with $\mathbf{S}$ diagonal and some $|S_{kk}| \neq 1$ is not a symmetry, because it changes the prior divergence. The centred shift $\mathbf{B}(\mathbf{w}_i - \bar{\mathbf{w}}_t)$, with $\bar{\mathbf{w}}_t$ the mean response of type t , and the realised prior shift $\mathbf{B}[m_\psi(\mathbf{c}_i, t_i) - m_\psi(\bar{\mathbf{c}}_t, t_i)]$, with $\bar{\mathbf{c}}_t$ its mean niche descriptor, each centred over genes, are unchanged by all three maps.

The fixed prior scale σ_w is what removes the rescaling: a learned constant σ_w would absorb it, and a learned conditional $\sigma_w(\mathbf{c}, t)$ could shrink the divergence by inflating the prior wherever it bites. Loadings are read only after centring each column over genes. The per-type translation holds only approximately, because the decoder sees $\mathbf{z}$ but not t and posteriors of different types overlap; it is a near-flat direction that the optimisation, not the likelihood, decides. Hence the magnitude $\|\mathbf{w}_i\|$, and any ranking of types by it, is a statement about the gauge; the gauge-free quantities are the two centred shifts. A “clean” per-cell profile must be decoded at a reference niche such as $\bar{\mathbf{c}}_t$, not at $\mathbf{w} = \mathbf{0}$, which would subtract the type offset along with the niche effect. Differences between two niches, which the relocation reads, are gauge-free. The allocation between $\mathbf{z}$ and $\mathbf{w}$ rests on the conditional invariance, the intrinsic path and $d_w < d_z$. For $\mathbf{w}$ given the

niche this is in the spirit of the auxiliary-variable setting of Khemakhem et al. [8], whose guarantee we do not establish for this likelihood; for $\mathbf{z}$ no comparable guarantee exists without the constraint [9], which the held-out probe verifies rather than assumes.

Proof of Proposition 1. Write $\mathbf{u}_i = a(\mathbf{z}_i) + \mathbf{B}\mathbf{w}_i$ for the logits of Eq. (3). The terms of J that involve a , $\mathbf{B}$, $\mathbf{w}$ or m_ψ are the two reconstruction terms, which are expectations of functions of the logits of cell i and, through $\bar{\rho}_i$, of its neighbours, and the prior divergence of $\mathbf{w}$. The $\mathbf{z}$ -divergence and the invariance term depend only on $q(\mathbf{z}_i)$ and t_i and are untouched by all three maps.

(a) $\mathbf{B}\mathbf{Q}^\top \mathbf{Q}\mathbf{w}_i = \mathbf{B}\mathbf{w}_i$, so every draw has the same logits, hence the same ρ_i , $\bar{\rho}_i$ and $\mathbf{p}_i$. Under the prior of Eq. (2), $\mathbf{Q}\mathbf{w}_i \sim \mathcal{N}(\mathbf{Q}m_\psi, \sigma_w^2 \mathbf{Q}\mathbf{Q}^\top) = \mathcal{N}(\mathbf{Q}m_\psi, \sigma_w^2 \mathbf{I})$, which is the mapped prior, so the distribution of the counts is unchanged. In J , the mapped posterior and the mapped prior are the images of the originals under $\mathbf{w} \mapsto \mathbf{Q}\mathbf{w}$, so both reconstruction terms, the second of which draws $\tilde{\mathbf{w}}_i$ from the prior, are unchanged, and the divergence between two distributions is invariant under a common bijection. For a signed permutation, $\mathbf{Q} \text{diag } \nu^2 \mathbf{Q}^\top$ is the diagonal matrix with the entries of ν^2 permuted.

(b) The logits become $\mathbf{u}_i + (c(\mathbf{z}_i) + \mathbf{b}^\top \mathbf{w}_i)\mathbf{1}$, a shift by a scalar that is the same for every gene, and the softmax is invariant under such a shift. Every clean composition is therefore unchanged for every draw, and no other term of J involves a or $\mathbf{B}$.

(c) For $\mathbf{z}$ in the support of $q(\mathbf{z}_i)$, $\tau(\mathbf{z}) = t_i$, so the logits become $a(\mathbf{z}) - \mathbf{B}\delta_{t_i} + \mathbf{B}(\mathbf{w} + \delta_{t_i}) = a(\mathbf{z}) + \mathbf{B}\mathbf{w}$, both for posterior draws, which are shifted with the posterior mean, and for prior draws $\tilde{\mathbf{w}}_i$, which are shifted with m_ψ . The same holds for every neighbour, so $\bar{\rho}_i$ is unchanged. The divergence between two Gaussians with the same covariances depends on their means only through their difference, and both means are shifted by δ_{t_i} .

Rescaling. With $\lambda_k = S_{kk}^2$, dimension k of the divergence between $\mathcal{N}(\mu_{w,i}, \text{diag } \nu_{w,i}^2)$ and $\mathcal{N}(m_\psi, \sigma_w^2 \mathbf{I})$ changes under the map by $\frac{1}{2}[(\lambda_k - 1)A_{ik} - \log \lambda_k]$ with $A_{ik} = (\nu_{w,ik}^2 + (\mu_{w,ik} - \psi_{ik})^2)/\sigma_w^2$, where ψ_{ik} is coordinate k of $m_\psi(\mathbf{c}_i, t_i)$. Because σ_w is fixed, for $\lambda_k \neq 1$ this vanishes only where $A_{ik} = \log \lambda_k / (\lambda_k - 1)$, not in general, so the map does not preserve J .

Readouts. Under (a) the differences $\mathbf{w}_i - \bar{\mathbf{w}}_t$ and $m_\psi(\mathbf{c}_i, t_i) - m_\psi(\bar{\mathbf{c}}_t, t_i)$ are multiplied by $\mathbf{Q}$ and $\mathbf{B}$ by $\mathbf{Q}^\top$ on the right, so their products are unchanged. Under (b) each product gains $\mathbf{1}\mathbf{b}^\top(\cdot)$, a constant over genes, which centring removes. Under (c) both terms of each difference are shifted by the same δ_t , and the contexts are unchanged. $\square$

S2 Implementation, Design Choices and Calibration

S2.1 Design Choices

The type as attention query. In the attention layer of Eq. (7) the destination feature enters the attention weights but not the aggregated values [11]. Querying with $\mathbf{z}_i$ would still open a selection channel: attention could weight neighbours by their similarity to $\mathbf{z}_i$, so that $\mathbf{c}_i$ mirrors the cell, m_ψ already knows it, and the response divergence collapses, with $\mathbf{w}_i$ measuring deviation from *self*. The query is therefore the cell’s type, at no cost to the prior, which receives t_i anyway; the sources are the neighbours’ types, chosen over types plus the neighbours’ posterior means because a three-seed ablation favoured them. The *mirror* diagnostic is the within-type R^2 of $\mathbf{z}_i$ regressed on $\mathbf{c}_i$ against a within-type permuted control; it must be ruled out before a vanishing response divergence is read as posterior collapse.

No per-edge geometry; Delaunay graph and Voronoi kernel. The attention uses no distances or face lengths, so that it cannot rediscover the spill-over kernel β and merge the two mechanisms. With the type as query and one-hot types as sources, the logit of edge ij depends on (t_i, t_j) alone, say $s_{t_i t_j}$, and per head

$$\mathbf{g}_i = \frac{\sum_k y_{ik} e^{s_{t_i k}} \mathbf{W}_{\text{src}} \mathbf{e}_k}{\sum_k y_{ik} e^{s_{t_i k}}}, \quad (\text{S3})$$

a type-specific reweighting of the composition $\mathbf{y}_i$ in which the degree cancels. The context thus depends on fractions, not counts; on a Delaunay graph the degree is nearly constant, so this costs little. Delaunay adjacency (Voronoi cells sharing a face) is built from centroids alone, with one parameter, the 40 μm prune.

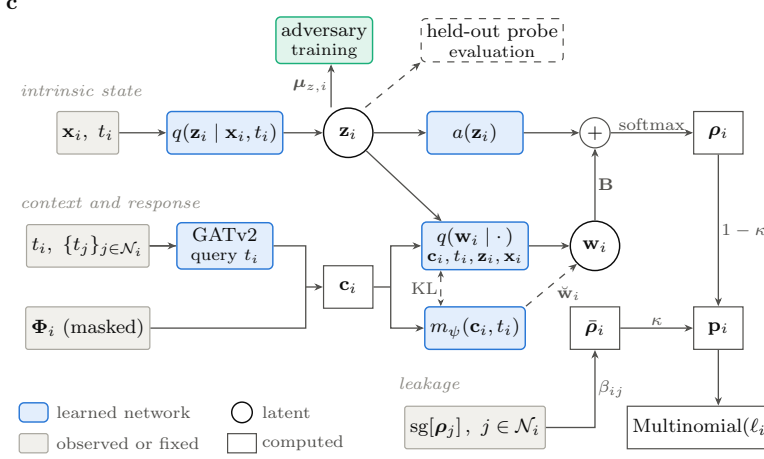


Fig. S1: The computation graph of DISCELL for one cell i . $\mathbf{z}_i$ is encoded from the cell’s own counts and type; $\mathbf{c}_i$ combines attention over the neighbours’ types, queried with the cell’s type, with Φ_i ; $\mathbf{w}_i$ has the prior $p(\mathbf{w}_i | \mathbf{c}_i, t_i)$ with mean m_ψ and a posterior that also sees the cell. ρ_i is mixed with the neighbours’ clean compositions, held under a stop-gradient (sg), through β and the fixed κ . The dashed $\tilde{\mathbf{w}}_i$ is the prior draw of the intrinsic path. The adversary acts on $\mu_{z,i}$ during training; the held-out probe grades it afterwards.

A discretised contact indicator would not avoid the confound: on these sections, segmented mostly (69 to 87% of cells) by an interior RNA stain, adjacent polygons rarely touch, and whether they do measures local density (Table S2). A contact kernel is zero on about half of the edges and leaves up to a third of the cells in the sparsest fifth of a section with no spill-over at all; the Voronoi-face kernel gives every connected cell the same budget κ . Edge features in the attention were not tested (Sec. S8).

Table S2: Segmented-polygon contact as a spill-over kernel against the Voronoi-face kernel β , on the model’s edge set; both use the same distance decay and row normalisation. *Touching*: adjacent polygons at zero distance; *no contact*: edges with no membrane within $1 \mu\text{m}$. *No spill-over*: connected cells whose contact-kernel row is all zero; *sparse/dense*: lowest/highest fifth of cells by centroids within $20 \mu\text{m}$. ρ : Spearman correlation over edges between a neighbour’s share of the row and the gap between the polygons (decay alone: -0.49 to -0.62). The Voronoi kernel has no zero-weight edges on any section. FF: fresh frozen.

Section	Edges (%)		No spill-over, contact (%)			$\rho(\text{share, gap})$	
	touching	no contact	all	sparse	dense	contact	β
Ovarian cancer (FFPE)	3.5	47	9.8	33.6	0.6	-0.82	-0.48
Lung cancer (FFPE)	2.7	57	11.1	27.4	0.7	-0.84	-0.55
Ovarian cancer (FF)	6.7	38	3.2	10.6	0.1	-0.80	-0.52
Lung TMA, section A	2.2	47	6.3	24.8	0.1	-0.84	-0.54
Lung TMA, section B	2.2	47	6.2	24.6	0.1	-0.84	-0.53

The image descriptor and its mask. Φ_i is the 384-dimensional embedding by KRONOS [13] (first release, ViT-S/16) of a $128 \mu\text{m}$ field (256 pixels) in the section’s four morphology channels. The channels map onto the model’s marker vocabulary only approximately: the membrane and αSMA channels are antibody mixes read as their named component, and 18S RNA is given an unused marker identity. The cell is removed by zeroing a disc of fixed radius centred on the crop anchor, not its own polygon, whose outline would reveal its type. A $25 \mu\text{m}$ disc leaves six cells of the primary section outside it (elongated smooth-muscle cells, whose descriptor is set to zero) and masks 12% of the field; at this density it also removes the first ring or two, so Φ_i describes the tissue beyond roughly $25 \mu\text{m}$. Table S3 shows what remains: the masked patch predicts the cell’s label less well than the neighbours’ labels alone, so what it knows about the cell is what its neighbourhood

implies. On the masked patch the first release does at least as well as KRONOS2 at half the embedding width, so we use it.

Table S3: What each mask leaves in the image embedding (primary section): macro one-versus-rest AUC of a multinomial logistic regression predicting the cell’s lineage label (11 classes) from the embedding, on a spatial split (1 mm tiles, cells within half a field of a tile edge dropped; 40,000 training and about 74,000 test cells). The model uses KRONOS with the disc zeroed (bold).

Input to the classifier	KRONOS	KRONOS2
Neighbours’ labels, no image		0.918
Whole patch	0.881	0.878
Disc over the cell zeroed	0.857	0.846
The cell alone	0.905	0.919
The cell alone, native resolution	0.923	–

A direct path from the counts into $q(\mathbf{w})$. The response posterior reads the cell’s counts, because $\mathbf{w} \perp \mathbf{x} \mid \mathbf{z}, \mathbf{c}$ is false by explaining-away and $\mathbf{z}$ is lossy in exactly the wrong direction: the invariance removes from it the niche-predictable variation that $\mathbf{w}$ exists to capture. Without the path, $\mathbf{w}_i$ collapses to the expected response given type and niche.

Conditional invariance and label granularity. The invariance of Eq. (9) is a constraint on the variational family, not a property of the generative model (Sec. S1.2). It removes from $\mathbf{z}$ only what the neighbour composition and the masked image predict given the type. Intrinsic variation that is spatially organised within a type, such as subclones in contiguous regions whose kin the masked image shows, is partly predictable from the niche, is moved into $\mathbf{w}$ and would be read as a response; on a carcinoma section this is the main case, not a corner case. The decomposition is therefore relative to the labels. We use lineage-level labels because finer or state-defined labels leave $\mathbf{z}$ little to carry and destroy the overlap that conditional comparisons need, and because t_i is itself a function of the counts, so conditioning on it conditions on a descendant of the niche near label boundaries [15]; marginal invariance instead collapses the agreement with type [1].

S2.2 Architecture and Hyperparameters

Table S4 lists every network and every fixed quantity as implemented. All multilayer perceptrons use rectified-linear hidden units.

S2.3 Calibration

Every weight was set once, on the primary section, and used unchanged on every section; Sec. S7.4 refits the alternatives. Table S5 records what was tuned, by which read, and where it was checked.

The invariance threshold is not met everywhere. During development, a closed-form Gaussian estimate of the conditional mutual information, evaluated on the posterior means with $\mathbf{v}_i = [\mathbf{y}_i^{(-K)}, \text{PC}_{12}(\Phi_i)]$,

$$\text{Pen} = \sum_t p(t) \cdot \frac{1}{2} [\log \det \hat{\Sigma}_{\mathbf{v}|t} + \log \det \hat{\Sigma}_{\mathbf{z}|t} - \log \det \hat{\Sigma}_{[\mathbf{z}, \mathbf{v}]|t}], \quad (\text{S4})$$

served in place of the adversary; it removes linear dependence only, and the nonlinear probe of a converged fit stayed above its floor, so the adversary replaced it. The development target for the adversary was a nonlinear residual of at most a quarter of that of the fit without the adversary, on both blocks and for both probe families. At the operating point the nonlinear probe exceeds that threshold on the composition block on three sections and on the image block on the TMA core; on the TMA core and the lung section the residual without the adversary is itself small, so a similar absolute residual is a larger fraction of it. The residual niche signal is therefore reported descriptively, in absolute terms and as a fraction of the fit without the adversary (Table S9), not as a pass or a fail.

Table S4: Networks and fixed quantities. G genes, K types, d_Φ image-embedding dimension.

Component	Input	Layers / value
Count encoding $q(\mathbf{z}_i \mathbf{x}_i, t_i)$	$\mathbf{x}_i$ count encoding $\oplus$ onehot(t_i)	$\lceil \log(1 + \mathbf{x}_i \tilde{\ell} / \ell_i) \rceil, \log \ell_i$, $\tilde{\ell}$ the median total $256 \rightarrow 256 \rightarrow 2d_z, d_z = 20$
Type query embed(t_i)	t_i	learned table, width K
Attention sources	neighbour j	onehot(t_j), width K
Attention layer	sources, query	one bipartite GATv2 layer, 4 heads $\times$ 32, averaged, LeakyReLU slope 0.2, no self-loops, no edge features
Context $\mathbf{c}_i$		attention output (32) $\oplus \Phi_i$ ($d_\Phi = 384$) $\oplus$ isolation flag (1)
Prior mean $m_\psi(\mathbf{c}_i, t_i)$	$\mathbf{c}_i \oplus$ onehot(t_i)	$64 \rightarrow d_w, d_w = 6$
$q(\mathbf{w}_i \cdot)$	$\mathbf{c}_i \oplus$ onehot(t_i) $\oplus \mathbf{z}_i \oplus$ count encoding	$256 \rightarrow 2d_w$
Decoder $a(\mathbf{z}_i)$	$\mathbf{z}_i$	$256 \rightarrow G$
Loadings $\mathbf{B}$	$\mathbf{w}_i$	linear $d_w \rightarrow G$, no bias
Adversary heads	$\mu_{z,i} \oplus$ onehot(t_i)	two heads $64 \rightarrow 64 \rightarrow K$ (composition) and $\rightarrow E_\Phi = K$ (image niche)
Image niche e_i^Φ	12 principal components of Φ	soft k -means memberships softmax($-d^2/T$), T the median nearest squared distance; projection fitted on a 50,000-cell subsample, k -means (four restarts) on all cells; frozen
Held-out probes	$\mu_{z,i} \oplus$ onehot(t_i)	ridge ($\lambda = 10^{-3}$) and perceptron $64 \rightarrow 64$ (300 Adam steps at 10^{-3} , batches of 2,048), per block against its own within-type permutation floor; fitted on at most 30,000 training-tile cells
Graph		Delaunay adjacency; Voronoi faces measured within $30 \mu\text{m}$ discs; prune at $40 \mu\text{m}$
Kernel β		$f_{ij} \exp(-d_{ij}/\tau)$, $\tau = 20 \mu\text{m}$, row-normalised over in-edges
Prior scale σ_w		1 (fixed)
Spill-over fraction κ		swept over $\{0, 0.05, 0.1, 0.2, 0.3, 0.4\}$; operating point 0.1
ω		1
α_z		$\frac{1}{2}/\bar{\ell}$, $\bar{\ell}$ the section's median total
α_w		0.1, ramped linearly from 0 over the first 30 epochs
α_a (adversary)		0.3; six head updates per model update at learning rate 2×10^{-3} ; composition weighted $\lambda_y = 3$
Tiles		recursive median split into rectangles of $\leq 4,096$ cells (2,048 on the TMA core); 15% of tiles held out
Optimiser		Adam [20], learning rate 10^{-3} , cosine annealing [21], gradient-norm clip 10; 500 epochs, patience 40, for final fits and sweep alike
Evaluation		every 5 epochs on posterior means; checkpoint accepted if held-out reconstruction improves and $\mathbf{z}$ -type agreement $\geq 0.9 \times$ its running maximum; none during the α_w ramp
Hardware		one NVIDIA GeForce RTX 4090 (24 GB) per fit

Table S5: What was calibrated, on which section, by which read. Primary: the ovarian FFPE section.

Setting	Set on	Read	Checked
$\alpha_z = \frac{1}{2}/\bar{\ell}$	primary	ladder $\{\frac{1}{4}, \frac{1}{2}, 1, 2\} \times 1/\bar{\ell}$, read on the diagnostics of Sec. 2.3; $\frac{1}{2}$ restores one charge of the $\mathbf{z}$ -divergence at $\omega = 1$ (Sec. S1.2)	–
$\alpha_w = 0.1$	primary	at lower weights $\mathbf{w}$ takes over type structure and the $\mathbf{z}$ -type agreement falls	$1/\bar{\ell}$ on primary and TMA core; $5/\bar{\ell}$, $10/\bar{\ell}$ on FF
$\alpha_a = 0.3$, head steps	primary	nonlinear held-out probe, without loss in $\mathbf{z}$ -type agreement or held-out reconstruction; head strength judged by the probe, never by the heads' loss [7]	doubled steps, width, ensemble
$\lambda_y = 3$	primary	composition residual 4.6% against 6.6% at equal weights	$\lambda_y = 1, 5$
$\omega = 1$	–	two bounds weighted equally	$\omega = 0$ on all four sections

S2.4 Training, Diagnostics and Seeds

The adversary’s image target e_i^Φ is a soft membership in $E_\Phi = K$ image niches from k -means on the leading principal components of Φ_i , never on the raw embedding, where distances are noise-dominated; $E_\Phi = K$ gives the two heads comparable difficulty. The composition head is weighted $\lambda_y = 3$, in the encoder term and in the heads’ loss, because composition is the channel of the spill-over confound.

The response divergence is ramped because without the ramp the conditional prior can fail to open: the posterior of $\mathbf{w}$ collapses within the first ten epochs onto a prior that ignores $\mathbf{c}$, shown by a divergence at zero and no information between $\mathbf{w}$ and the niche on held-out cells. Such dead-channel fits were discarded and re-seeded; none is among the final fits. The running maximum of the type agreement ratchets upward, so a checkpoint that buys reconstruction with type structure is rejected. Held-out cells are entire spatial tiles, because spatial autocorrelation passes the answer across a random split [22]. Every fit is monitored by the mirror R^2 (Sec. S2.1), the response divergence per dimension, the held-out probe (Eq. (11)), the $\mathbf{z}$ -type agreement, and two checks that $\mathbf{z}$ has not collapsed onto the label: $I(\mathbf{z}; t)/H(t)$ is 0.82 to 0.95 on the final fits, the within-type share of the variance of μ_z 0.60 to 0.72, and decoding from the cell’s own $\mathbf{z}$ rather than its type’s mean gains 0.06 to 0.17 nats per count (Table S23). The objective is not unimodal in the seed: repeated fits of one configuration can fall into different basins. Each seed also redraws the held-out split, the image principal components and the image-niche clustering. No seed is selected: figures use one seed fixed in advance, tables the mean with the range over seeds. A spill-over-subtracted encoder input, an attention sink and coupled weight decay are implemented but off in every reported fit.

S2.5 Tiles and the Two-Hop Halo

Because $\bar{\rho}_i$ needs ρ_j , hence $\mathbf{w}_j$, hence $\mathbf{c}_j$, a training step for cell i reaches two graph hops. The neighbours’ responses are posterior draws from the same batched pass; dropping them would strip the local programme from $\bar{\rho}_i$ and let $\mathbf{w}_i$ absorb it, inflating apparent spatial effects. The neighbours’ compositions enter under a stop-gradient, as in resolVI [12], which closes the feedback path through which ρ_j would be trained to absorb cell i ’s residual. Batches are contiguous spatial tiles, whose halo is shared: a tile’s cells are the seeds, on which the loss is evaluated; their neighbours (ring one) get context, response and clean composition; ring two contributes only types. Both rings are read from the kernel’s own edge list, never from a distance query, so the halo and β cannot disagree. One consequence: the ring of a training tile can contain cells of a held-out tile, whose counts then enter training as stop-gradient inputs with no loss. On our sections 0.8 to 1.9% of training seeds have a held-out neighbour, and 4.8 to 10% of held-out cells enter some training step in this way; within two hops, where only the type is used, the share of seeds is 1.8 to 4%. We state the exposure rather than add a buffer, which would remove the same seeds.

S3 Data, Labels and Comparison Methods

S3.1 Sections

Table S6 lists the sections of Sec. 3.1, and Fig. S2 shows them. The lineage classes include Unassigned. On the ovarian FFPE section it also holds a low-depth class of mixed lineage that the curated annotation had named as a tumour subtype, which is why its share is the largest.

S3.2 Lineage Labels

The fitted labels are lineage-level on every section (Sec. 3.1): states (proliferating, activated, hypoxic) and locations (tumour- or stroma-associated) are merged into their lineage; lineages and developmentally distinct sub-lineages are kept. Mappings were fixed before any model was refitted. On the ovarian FFPE section the eighteen curated classes become twelve (Table S7). The class named as a SOX2-OT⁺ tumour subtype is mostly not tumour (a median of 27 transcripts against 333 for tumour cells, three quarters with no SOX2-OT count, a pooled profile mostly of macrophage expression) and is folded into Unassigned. The class named as malignant cells lining a cyst expresses mesothelial markers (CALB2, PRG4, BNC1, PDPN, UPK3B) and lacks the tumour markers (EPCAM, PAX8, ESR1), so it becomes its own lineage. The tumour- and

Table S6: The sections. Median counts are per cell over the whole section. The TMA core and its serial section are one core of the microarray on two slides; no model is fitted to the serial section.

Section	Role	Cells	Genes	Lineage classes	Unassigned (%)	Median counts
Ovarian cancer (FFPE)	primary	407,120	5,101	12	10.3	178
Lung cancer (FFPE)	fitted	278,324	5,001	20	0.1	242
Ovarian cancer (FF)	fitted	1,157,637	5,001	10	4.3	1,401
Lung TMA core	fitted	69,422	5,001	30	4.2	279
Lung TMA core, serial section	held out	70,757	5,001	30	4.8	253

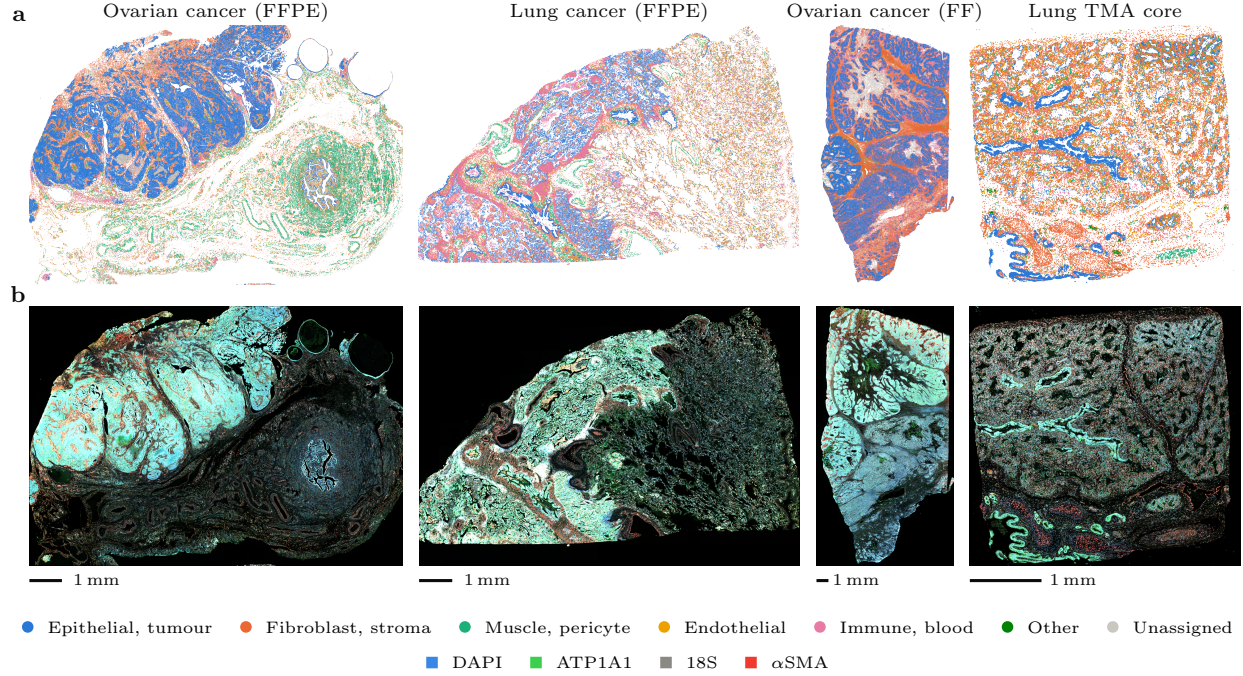


Fig. S2: The four fitted sections. **a**, Every cell at its centroid, coloured by its lineage label, with the labels grouped into six families for display only; the model is fitted with the finer lineage classes of Table S6. **b**, The morphology image of the same section, the four channels the image descriptor reads, shown as one colour composite (Sec. S2.1). The TMA panel is the fitted core only. Each section is shown at its own scale; the bars are 1 mm.

stroma-associated fibroblasts are merged, since the cancer-associated fibroblast programme that separates them is a state; the two endothelial classes are merged too, although what separates them (a tip-cell state against a weakly venous identity) is less clear. On the lung TMA the thirty-five curated classes become thirty: the state classes are reassigned cell by cell to the sub-lineage their markers indicate (right for about three cells in four), and the myofibroblasts are split about evenly between alveolar and adventitial fibroblasts. On the lung FFPE and fresh-frozen sections each unsupervised cluster is assigned the lineage its markers indicate (on the TMA's vocabulary for lung); immune clusters next to tumour that carry some tumour signal keep their immune lineage, since that signal is the spill-over the model absorbs. The cluster-to-lineage mappings accompany the code.

S3.3 Baselines

Table S8 gives the configuration of each comparison method. Every method runs at its published defaults, on our pruned Delaunay graph where it accepts one (resolVI and MintFlow build their own), and reads the lineage labels if it reads a label (SIMVI does not). Every comparison is on whole sections; a method that

Table S7: Lineage labels on the ovarian FFPE section: the curated classes merged into each lineage.

Lineage	Curated classes
Ciliated epithelial cells	Ciliated Epithelial Cells
Endothelial cells	Stromal Associated Endothelial Cells, Tumor Associated Endothelial Cells
Fallopian tube epithelium	Fallopian Tube Epithelium
Fibroblasts	Stromal Associated Fibroblasts, Tumor Associated Fibroblasts
Macrophages	Macrophages
Mesothelial-like cyst lining	Malignant Cells Lining Cyst
Pericytes	Pericytes
Smooth muscle cells	Smooth Muscle Cells
T/NK cells	T and NK Cells
Tumour	Inflammatory Tumor Cells, Proliferative Tumor Cells, Tumor Cells, VEGFA+ Tumor Cells
Unassigned	SOX2-OT+ Tumor Cells, Unassigned
Urothelial-like cells	Urothelial-like Cells

cannot be fitted to a whole section is reported as such, with the measured reason, never run on a window. Cellina removes a spatial-domain label adversarially; our sections carry none, so its main row runs without the adversary. It also runs with the adversary given our niche label (clusters of neighbour composition, $K = 10$, fitted on training cells), which matches what our probe grades and is its best case on the probe, not its published setting; on its own neighbour graph, on the TMA core and its serial section; and, for Table S17, refitted on the training tiles and decoded at its posterior mean. Tables S9 and S10 give every method’s values. MintFlow’s reconstruction is not reported: an export error in our pipeline passed MintFlow’s split of each cell’s observed counts, which sums back to those counts, in place of a decode, so the stored read scored each cell against its own counts. MintFlow is being refitted with a corrected export, and all its reads will come from the new fits [PENDING RUNS: MintFlow refits with the corrected export].

Reconstruction on one basis. The methods differ in likelihood: DISCELL’s is a multinomial given the cell’s total, the comparison methods use their own count families, and SIMVI has no count decoder. The battery therefore scores every method the same way: its decoded expected counts are renormalised per cell to a probability vector $\mathbf{p}_i$, and the read is the held-out per-count log-likelihood $\ell_i^{-1} \sum_g x_{ig} \log p_{ig}$, averaged over cells. A method’s dispersion or zero-inflation parameters do not enter, so the read compares decoded mean profiles, not the methods’ own likelihoods.

Cellina variants. Given our niche label as its domain, Cellina’s composition residual does not consistently fall: the nonlinear residual falls on three sections of five and rises on lung and the TMA core, by at most 2.0 points either way; the linear one rises on four of five; NMI with type and cycle R^2 fall on every section. On its own neighbour graph Cellina keeps less cycle state than DISCELL (cycle R^2 0.39 on the TMA core and 0.31 on its serial section, against DISCELL’s 0.48–0.51 and 0.49–0.50) at a nonlinear composition residual similar to DISCELL’s on the core (4.2% against 3.5–4.8%) and higher on the serial section (8.3% against 4.0–4.8%); against Cellina on our graph, its residual is slightly higher and its cycle R^2 higher (Tables S9 and S10).

S3.4 Assets and Licences

The ovarian FFPE, lung FFPE and fresh-frozen ovarian sections are public datasets of 10x Genomics [23–25], released under the Creative Commons Attribution 4.0 International licence (CC BY 4.0). The TMA core and its serial section are two public samples of GEO series GSE315411 [26]. The image model KRONOS and its successor KRONOS2 [13] are released under CC BY-NC-ND 4.0, for non-commercial academic research and after registration; we use them as distributed by their authors, for that purpose, and redistribute neither. The comparison methods are released under the BSD 3-Clause licence: scvi-tools, which contains resolVI, SIMVI, MintFlow and Cellina. A code repository accompanies the submission as a clean, standalone implementation

Table S8: Comparison methods as run. Versions are the package versions used. *Latents*: the latent each method’s intrinsic reads are taken from, and its context latent. *Coverage*: the sections with a completed whole-section fit, and the state of the others.

Method	Version	Graph	Label	Latents: intrinsic / context	Budget and coverage
resolVI	scvi-tools 1.5.1	own, 10 nearest neighbours	lineage	its single cell latent / none (neighbour spill-over is a mixture, not a latent)	50 epochs; every section; on the serial section refitted there, since the transfer from the core failed
SIMVI	simvi 0.1.2 (scvi-tools 0.16.1)	ours	not used	intrinsic latent / spatial latent	package default; the TMA core, and the serial section fitted on itself; out of GPU memory (24 GB card) on the other three sections, measured reasons in Table S10
MintFlow	mintflow 0.3.0	own Delaunay graph	lineage	intrinsic latent / microenvironment latent	width 600, 50 epochs; the TMA core, transferred to its serial section, and the lung FFPE and ovarian FFPE sections; out of host memory (125 GB host) before training on the fresh-frozen section (1.16 M cells), measured reason in Table S10
Cellina	cellina 1.1.1 (scvi-tools 1.2.0)	ours	lineage	intrinsic latent / microenvironment latent <i>s</i>	package default, no domain adversary; every section, transferred from the core to its serial section
Cellina, niche domain	as Cellina	ours	lineage; niche label as domain	as Cellina	as Cellina; every section
Cellina, own graph	as Cellina	own	lineage	as Cellina	as Cellina; the TMA core and its serial section
Cellina, counterfactual	as Cellina	ours	lineage	as Cellina	refitted on training tiles only; the four fitted sections

of DISCELL; it does not contain KRONOS or its weights, which are obtained from their authors under their licence. No new data are released.

S3.5 What the Model Needs

Every fit in this paper is to a section profiled with Xenium Prime 5K, from whose output the model uses five inputs: the counts per cell; one position per cell (the centroid of the segmented cell), from which the graph, the kernel, the tiles and the image fields are built (Sec. S2.1); the pixel size; a type label per cell, from a curated annotation or from clusters mapped to lineages (Sec. S3.2); and the four-channel morphology image. Segmented polygons are not required; they served only to choose the radius of the disc that removes the cell from its image field, which a new tissue or platform would need to recheck (Sec. S2.1), and for the graph and mask comparisons. Transcript coordinates, nucleus boundaries and the post-run H&E image are not used; cell-cycle genes enter a diagnostic only (Sec. 2.3). Implicitly, counts must be assigned to segmented cells at single-cell resolution, since the spill-over term models transcripts misassigned between neighbouring cells; one fit is to one contiguous section; and the multinomial given each cell’s total suits the hundreds to about a thousand counts per cell of in situ data (Sec. 2.1). No other assay has been tested.

Table S9: The held-out probe on the final fits, per section. For each block and probe, the share of the block’s within-type variance that the probe explains from the intrinsic latent beyond the permutation floor ($1 - e^{-2\text{excess}}$, in %): for the model trained without the adversary ($\alpha_a = 0$, two seeds), with it (the final fits, three seeds), and for the comparison methods on the same section. The *fraction left* is the excess with the adversary as a fraction of the mean excess without it: 0 when the adversary removes everything the probe can find, 1 when it removes nothing. Ranges are over seeds; Unassigned cells are not targets. Arrows give the preferred direction; bold marks the best value per section and probe block, among DISCELL with the adversary (by its mean over seeds) and the comparison methods. The serial section of the TMA core is graded with our models fitted on the core; there, the Cellina rows and MintFlow are the core’s models transferred, and resolVI and SIMVI are fitted on the serial section itself. *Cellina, niche domain*: Cellina with its domain adversary given our niche label (clusters of neighbour composition), which matches what this probe grades, so it is Cellina’s best case on the probe, not its published setting. *Cellina, own graph*: Cellina on its own neighbour graph instead of ours. SIMVI on ovarian FFPE could not be run on the resources currently available; SIMVI on lung FFPE could not be run on the resources currently available; SIMVI on ovarian FF could not be run on the resources currently available; MintFlow on ovarian FF could not be run on the resources currently available (-). -: not run on that section.

	Ovarian FFPE	Lung FFPE	Ovarian FF	TMA core	TMA serial
<i>Composition, MLP probe</i> ↓					
without the adversary (%)	16.1–16.2	10.7–11.5	27.6–28.0	6.4–6.9	6.6–6.9
with the adversary (%)	4.0–5.5	4.2–5.0	4.4–5.0	3.5–4.8	4.0–4.8
fraction left ↓	0.23–0.32	0.36–0.43	0.14–0.16	0.52–0.71	0.59–0.70
resolVI, Cellina (%)	17.4, 13.9	11.2, 7.9	23.6, 22.9	5.4, 3.5	7.3, 7.6
MintFlow, SIMVI (%)	3.0 , –	1.1 , –	–	0.3 , 4.9	1.0 , 6.8
Cellina, niche domain (%)	13.4	8.5	21.6	5.5	7.4
Cellina, own graph (%)	–	–	–	4.2	8.3
<i>Image, MLP probe</i> ↓					
without the adversary (%)	12.6–15.8	8.8–9.8	27.1–28.7	6.0–8.4	5.6–7.2
with the adversary (%)	2.2–4.0	1.4–2.1	4.7–5.2	2.3–3.3	2.5–3.0
fraction left ↓	0.15–0.26	0.14–0.21	0.15–0.16	0.31–0.45	0.38–0.46
resolVI, Cellina (%)	13.1, 9.4	10.1, 4.2	23.4, 12.9	7.2, 3.9	7.2, 2.8
MintFlow, SIMVI (%)	2.8 , –	0.7 , –	–	0.8 , 8.2	0.6 , 7.9
Cellina, niche domain (%)	9.7	4.7	14.8	5.3	4.1
Cellina, own graph (%)	–	–	–	4.7	4.4
<i>Composition, ridge probe</i> ↓					
without the adversary (%)	5.6–6.7	3.7–3.9	16.6–16.8	2.4–2.6	2.5–2.6
with the adversary (%)	1.0–1.1	0.6–0.8	1.3–1.6	0.4–0.5	0.8–1.0
fraction left ↓	0.15–0.18	0.15–0.20	0.07–0.09	0.16–0.20	0.31–0.42
resolVI, Cellina (%)	7.4, 8.1	4.5, 4.2	12.2, 18.2	2.4, 2.1	2.6, 3.0
MintFlow, SIMVI (%)	4.0, –	1.0, –	–	0.9, 3.0	1.1, 3.0
Cellina, niche domain (%)	8.6	4.8	17.2	3.6	3.8
Cellina, own graph (%)	–	–	–	2.6	3.5
<i>Image, ridge probe</i> ↓					
without the adversary (%)	5.9–7.9	3.4–3.5	17.4–19.1	2.7–3.2	1.9–2.7
with the adversary (%)	0.6–1.4	0.2–0.5	2.1–2.6	0.5–0.7	0.5–0.8
fraction left ↓	0.08–0.19	0.06–0.15	0.11–0.13	0.18–0.23	0.22–0.36
resolVI, Cellina (%)	6.4, 6.7	3.9, 2.4	12.5, 10.4	2.5, 2.1	2.5, 1.9
MintFlow, SIMVI (%)	2.9, –	0.7, –	–	0.8, 3.7	0.7, 3.3
Cellina, niche domain (%)	7.3	3.2	12.2	3.4	2.8
Cellina, own graph (%)	–	–	–	2.8	2.7

Table S10: One battery for every method, at lineage labels, on the held-out cells of each section, with Unassigned cells not targets. NMI of the intrinsic latent with the type; *Ridge*, *MLP*: the held-out excess of the ridge and MLP probes of neighbour composition as a fraction of that of DISCELL without the adversary (the fraction left of Table S9); mirror R^2 ; cycle R^2 of the intrinsic latent on the top-decile cycling set; held-out reconstruction in nats per count. For DISCELL, the mean over three seeds and the range. Arrows give the preferred direction; bold marks the best value per section, for DISCELL by its mean. DISCELL’s held-out tiles are excluded from its training loss. The comparison methods are fitted as their software is designed, on every cell of the section with at least five counts, held-out tiles included, and are scored on our held-out cells: resolVI and MintFlow train on all of them, and Cellina and SIMVI on a random nine tenths of the cells (the rest serve their own early stopping), so their reads, reconstruction above all, are not held-out reads. On the serial section, resolVI and SIMVI are fitted on that section itself, so theirs are not held-out reads either; the Cellina rows, MintFlow and DISCELL are the core’s models transferred. *Cellina, own graph*: Cellina on its own neighbour graph instead of ours. DISCELL’s rows are the battery’s own reads, on the cell set shared with the comparison methods, and so differ slightly from Table S11. ^bMintFlow’s reconstruction is not reported: the stored value came from an export error in our pipeline that scored each cell against its own counts; refits with a corrected export are running. ^cSIMVI has no count decoder. ^dCellina, niche domain: Cellina with its domain adversary given our niche label (clusters of neighbour composition, fitted on training cells) in place of the tissue regions of its own paper, which our sections do not have. The label matches what the composition probe grades, so this row is Cellina’s best case on the probe, not its published setting. ^eMeasured reason: SIMVI, ovarian FFPE: out of GPU memory on a 24 GB card (a further 7.67 GiB requested with 15.87 GiB in use); SIMVI, lung FFPE: out of GPU memory on a 24 GB card (a further 5.16 GiB requested with 21.27 GiB in use); SIMVI, ovarian FF: out of GPU memory on a 24 GB card (a further 40.00 MiB requested with 23.47 GiB in use); MintFlow, ovarian FF: an earlier attempt was stopped for lack of host memory before training (1,157,343 cells, 5,001 genes; an estimated 92 GB needed on a 125 GB host); a retry needs 110 GB free and 102 GB were available. -: not applicable.

Section	Method	NMI $\uparrow$	Ridge $\downarrow$	MLP $\downarrow$	Mirror $R^2 \downarrow$	Cycle $R^2 \uparrow$	Recon. $\uparrow$
Ovarian FFPE	DISCELL	0.708	0.16	0.27	0.049	0.519	-7.192
	range	0.694–0.720	0.15–0.18	0.23–0.32	0.044–0.051	0.504–0.548	-7.231 to -7.148
	resolVI	0.711	1.22	1.09	0.293	0.559	-7.231
	SIMVI	could not be run on the resources currently available ^e					
	MintFlow	0.962	0.65	0.17	0.053	0.062	n/r ^b
	Cellina	0.850	1.35	0.85	0.103	0.566	-7.231
	Cellina, niche ^d	0.653	1.43	0.82	0.175	0.329	-7.278
Lung FFPE	DISCELL	0.733	0.17	0.39	0.026	0.186	-7.254
	range	0.728–0.741	0.15–0.20	0.36–0.43	0.025–0.027	0.177–0.195	-7.267 to -7.243
	resolVI	0.845	1.18	1.01	0.153	0.147	-7.282
	SIMVI	could not be run on the resources currently available ^e					
	MintFlow	0.985	0.26	0.09	0.039	0.006	n/r ^b
	Cellina	0.937	1.10	0.70	0.047	0.311	-7.276
	Cellina, niche ^d	0.811	1.27	0.76	0.048	0.180	-7.312
Ovarian FF	DISCELL	0.688	0.08	0.15	0.091	0.757	-7.320
	range	0.667–0.722	0.07–0.09	0.14–0.16	0.086–0.094	0.744–0.768	-7.329 to -7.312
	resolVI	0.687	0.71	0.83	0.400	0.489	-7.319
	SIMVI	could not be run on the resources currently available ^e					
	MintFlow	could not be run on the resources currently available ^e					
	Cellina	0.823	1.10	0.80	0.142	0.675	-7.309
	Cellina, niche ^d	0.587	1.03	0.75	0.138	0.614	-7.315
TMA core	DISCELL	0.618	0.18	0.60	0.028	0.497	-7.219
	range	0.611–0.627	0.16–0.20	0.52–0.71	0.028–0.028	0.476–0.509	-7.223 to -7.213
	resolVI	0.721	0.98	0.80	0.177	0.336	-7.214
	SIMVI	0.504	1.21	0.73	0.084	0.567	n/a ^c
	MintFlow	0.895	0.35	0.04	0.031	0.054	n/r ^b
	Cellina	0.841	0.83	0.51	0.046	0.285	-7.218
	Cellina, niche ^d	0.769	1.47	0.81	0.130	0.233	-7.255
TMA serial	Cellina, own graph	0.823	1.06	0.63	0.063	0.388	-7.216
	DISCELL	0.607	0.37	0.64	0.039	0.495	-7.237
	range	0.601–0.616	0.31–0.42	0.59–0.70	0.039–0.040	0.494–0.496	-7.244 to -7.232
	resolVI	0.728	1.04 ¹³	1.08	0.173	0.344	-7.223
	SIMVI	0.486	1.21	1.01	0.081	0.631	n/a ^c
	MintFlow	0.911	0.45	0.14	0.035	0.056	n/r ^b
	Cellina	0.644	1.20	1.13	0.058	0.240	-7.274

S4 Readout Definitions

S4.1 Readouts

Every readout is taken at the accepted checkpoint of each fit. Unassigned cells are never targets but remain neighbours. A readout is the mean over three seeds with the range; unless stated otherwise its interval is from a bootstrap over 200 μm squares (tiles) of the section (1,000 draws, conditional on the fitted model and probes), reported as the envelope of the seeds' intervals (Table S12). Table S11 gives the same reads without intervals.

Niches. Where a readout needs discrete niches, they are the ten clusters of a k -means clustering of the neighbour composition $\mathbf{y}_i$ of connected cells; isolated cells have none.

Diagnostics. The *mirror* R^2 is the pooled R^2 of a linear regression of μ_z on the context $\mathbf{c}_i$, both centred per type, on up to 30,000 cells (Sec. 2.3). The response's information about the niche, $I(\text{niche}; \mathbf{w})$, is a nearest-neighbour estimate [31] of the mutual information between the niche label and μ_w on up to 20,000 held-out cells, in nats, minus its mean over five permutations of the label within type. The *cycle scores* are the S and G2/M scores of the gene sets of Tirosh et al. [32] intersected with the panel: each the mean depth-normalised log expression of the set minus that of expression-matched control genes, from the cell's own segmented counts [33]. They are a proxy for the cycle, not a measurement of it: the counts also hold transcripts spilled over from neighbours and expression induced by the environment, and both enter the scores (Sec. 3.3). The cycling set is the tenth of held-out cells with the highest summed score, fixed per section for every method, plus the training cells scoring above its lowest member. The *cycle* R^2 of a latent (μ_z , or μ_w for the cycle asymmetry of Sec. S7.1) is that of a ridge regression of both scores, centred per type within the set, on the latent, fitted on the set's training cells and scored on its held-out cells; the linear reference regresses on the leading 50 principal components of the log-normalised counts.

Relocation. A panel is a cell type and a pair of niches A and B , each with at least 500 cells supplying model quantities and 100 cells of the type to score; tiles are dealt into five spatial folds, one scored and four supplying every model quantity. Per gene, the observed shift is the difference between the two niches in the logarithm of the type's mean depth-normalised counts. The prediction holds niche A 's intrinsic rate fixed and adds two channels: the *programme part*, $\mathbf{B}$ times the difference between the mean prior means $m_\psi(\mathbf{c}_i, t)$ of the type's cells in B and in A , and the *spill-over part*, the change in the logarithm of the mixture of Eq. (5) when A 's mean spill-over influx is replaced by B 's at the fit's κ . No cell's own posterior response enters. A panel's *score* is the R^2 of the prediction against the observed shift, both centred over the genes expressed in both niches; the same score is computed for each part alone. The *noise ceiling* is the Spearman–Brown reliability of the observed shift from two random halves of the scored cells, averaged over four splits. Panels enter when their niches differ by more than three standard deviations of composition along the line between them; a panel is *trusted* when its ceiling is at least 0.5 with at least 100 genes scored. The *fraction of the ceiling* is the mean score over the trusted panels divided by their mean ceiling; over all panels the ratio divides by near-zero ceilings and can exceed one, so it is reported only in Table S18. The parts of Table S17 are the same ratio for each part alone; they are separate scores and do not add up. The breakdown contrast of Sec. S7.1 uses the mean score over all panels, without the trust filter: that of the full prediction minus that of the spill-over part alone (and, as a trajectory, minus that of the programme part alone, which is zero at $\kappa = 0$ by construction). The interval recomputes shifts, scores and ceilings on half of the tiles drawn without replacement and rescales the spread to the full section. On planted sections with tile-clustered noise its coverage is 0.94 to 0.96 at ceilings of 0.82 and 0.85, but 0.78 to 0.88 at ceilings of 0.52 to 0.63, just above the trusted threshold, so at the ceilings of real panels it is closer to an 80% than a 95% interval. Splitting the reliability halves by tiles instead of cells raises the trusted fraction by 0.01 to 0.02 on the seed mean of the ovarian FFPE, lung and fresh-frozen sections (-0.03 to $+0.05$ in single fits) and leaves the TMA core with no trusted panel; with the cell split, the TMA core has a trusted panel on two of its three seeds.

Single-cell read and twin read. On the same panels, without the trust filter, up to 2,000 held-out cells of the type in A are decoded from their own μ_z at B 's mean prior response and influx (the moved cloud) and

Table S11: Model quality on the four sections and the held-out serial section of the TMA core: the mean over three seeds, with the range over seeds in brackets; the 95% intervals are in Table S12. Reconstruction is the held-out log-likelihood in nats per count. Relocation is the fraction of the noise ceiling (cell-split reliability) recovered by the mean relocation read, over the panels in the trusted tier ([†]seeds with a trusted panel: TMA core, 2 of 3). The same read scored on held-out tiles only is a sensitivity row (Table S18). Cycle R^2 is read on the top decile of the S and G2M score among held-out cells. I(niche; $\mathbf{w}$) is the excess over a within-type permutation floor, in nats. The atlas cosine compares the first programme axis between seeds. The serial section is read through the models fitted on the core (–: not read there). Arrows give the preferred direction; ≈ 0 : expected to be near zero. Unassigned cells are not targets of any read.

Read	Ovarian FFPE	Lung FFPE	Ovarian FF	TMA core	TMA serial
Reconstruction $\uparrow$	–7.193 (–7.232 to –7.148)	–7.254 (–7.267 to –7.243)	–7.320 (–7.329 to –7.312)	–7.219 (–7.223 to –7.213)	–7.237 (–7.244 to –7.232)
NMI of $\mathbf{z}$ with the type $\uparrow$	0.713 (0.701–0.723)	0.736 (0.733–0.739)	0.697 (0.673–0.719)	0.621 (0.617–0.627)	0.611 (0.606–0.614)
Mirror R^2 $\downarrow$	0.048 (0.044–0.051)	0.026 (0.025–0.027)	0.091 (0.084–0.094)	0.028 (0.028–0.028)	0.039 (0.039–0.039)
Cycle R^2 of $\mathbf{z}$ $\uparrow$	0.519 (0.504–0.548)	0.186 (0.177–0.195)	0.757 (0.744–0.768)	0.497 (0.476–0.509)	0.495 (0.494–0.496)
Cycle R^2 of $\mathbf{w}$ (≈ 0)	0.006 (0.003–0.009)	0.003 (0.001–0.006)	0.001 (0.000–0.002)	0.000 (–0.000 to 0.002)	–0.001 (–0.004 to 0.002)
I(niche; $\mathbf{w}$) excess $\uparrow$	0.47 (0.45–0.50)	0.59 (0.58–0.62)	0.60 (0.58–0.62)	0.48 (0.42–0.52)	–
Relocation $\uparrow$	0.65 (0.63–0.67)	0.72 (0.68–0.75)	0.67 (0.65–0.68)	0.73 [†] (0.71–0.75)	–
Atlas cross-seed cosine $\uparrow$	0.97 (0.97–0.98)	0.84 (0.80–0.90)	0.82 (0.76–0.90)	0.91 (0.89–0.92)	–

at A ’s (the cloud left in place); the type’s held-out cells in B , decoded at their own posterior means, context and influx, are the target. The *single-cell read* (internally Read A) is the gap closed, $(U - T)/(U - F)$, where T and U are the maximum mean discrepancies [34] of the moved cloud and of the cloud left in place from the target, under a Gaussian kernel on square-root profiles, and F that between two random halves of the target; it is clipped to $[-1, 1]$ and summarised by its median over panels. Its *type-mean reference* is the same gap closed with the moved cloud replaced by copies of the target’s mean profile; the breakdown contrast is the single-cell read minus this reference. Its interval resamples the tiles of the source cells. The *twin read* (internally the twin margin) matches each target cell to the moved source cell nearest to it in μ_z ; with d_{twin} and d_{rand} the median Hellinger distances from the target cells’ own decodes to their matches and to random moved source cells, a panel’s margin is $(d_{\text{rand}} - d_{\text{twin}})/d_{\text{rand}}$, summarised by its median over panels; its interval resamples the tiles of the target cells. Because the decode is continuous in $\mathbf{z}$, part of this margin follows from the construction.

Response programmes. The programmes are the principal directions of the within-type-centred response that carry at least 1% of its variance, rotated by varimax [35] in gene space; their number is the effective rank. The cross-seed cosine compares the first programme’s gene signature between seeds; the programme overlap compares the gene subspaces two fits’ programmes span. Localisation compares each programme’s absolute loadings on four Gene Ontology cellular-component sets [36, 37] with the rest of the panel (Mann–Whitney, Benjamini–Hochberg within a fit [38]; effect: rank-biserial correlation).

External criteria. *Signalling-gene share*, after MintFlow [39]: on the relocation panels each gene’s observed shift is split into response part, spill-over part and remainder, and each gene’s share of each part is accumulated over panels; ligand and receptor genes of a curated database [40] are compared with the other genes by a Mann–Whitney test, and the breakdown contrast is the rank-biserial effect for the response share (interval by half-tile subsampling). *Niche information of both latents*: I(niche; $\cdot$) as above for μ_z and μ_w , also by a neural estimator [41]. *True and false axis*, after SIMVI [42], on the sections with tumour cells (the TMA sections have none): cells are banded by the fraction of tumour cells among their 50 nearest neighbours, in six bands. For each gene and type, the response-predicted shift per band, from the prior means of the type’s cells, is correlated by Kendall’s τ with the band index (the true axis) and with equal-count bins of the in-plane coordinate least correlated with the band index (the false axis); the breakdown contrast is the mean $|\tau|$ over all gene–type pairs on the true axis minus that on the false axis. The same is done for shifts predicted from the intrinsic state and from depth, and for the observed shift. The false axis is not orthogonal to the bands: both in-plane coordinates correlate with the band index at about 0.41.

Table S12: The reads of Table S11 with their intervals, on the four sections and the held-out serial section of the TMA core: mean over three seeds, the range over seeds and, where available, the 200 μm tile-bootstrap 95% interval (lowest lower and highest upper bound over the seeds). Reconstruction is the held-out log-likelihood in nats per count. Relocation is the fraction of the noise ceiling (cell-split reliability) recovered by the mean relocation read, over the panels in the trusted tier ([†]seeds with a trusted panel: TMA core, 2 of 3); its interval is from half-tile subsampling. The same read scored on held-out tiles only is a sensitivity row (Table S18). Cycle R^2 is read on the top decile of the S and G2M score among held-out cells. I(niche; $\mathbf{w}$) is the excess over a within-type permutation floor, in nats. The atlas cosine compares the first programme axis between seeds. The serial section is read through the models fitted on the core (–: not read there). Arrows give the preferred direction; ≈ 0 : expected to be near zero. Unassigned cells are not targets of any read.

Read	Ovarian FFPE	Lung FFPE	Ovarian FF	TMA core	TMA serial
Reconstruction $\uparrow$	–7.193	–7.254	–7.320	–7.219	–7.237
range over seeds	–7.232 to –7.148	–7.267 to –7.243	–7.329 to –7.312	–7.223 to –7.213	–7.244 to –7.232
95% interval	[–7.246, –7.125]	[–7.280, –7.230]	[–7.334, –7.307]	[–7.240, –7.200]	–
NMI of $\mathbf{z}$ with the type $\uparrow$	0.713	0.736	0.697	0.621	0.611
range over seeds	0.701–0.723	0.733–0.739	0.673–0.719	0.617–0.627	0.606–0.614
95% interval	[0.695, 0.729]	[0.728, 0.746]	[0.668, 0.725]	[0.610, 0.637]	–
Mirror $R^2 \downarrow$	0.048	0.026	0.091	0.028	0.039
range over seeds	0.044–0.051	0.025–0.027	0.084–0.094	0.028–0.028	0.039–0.039
Cycle R^2 of $\mathbf{z} \uparrow$	0.519	0.186	0.757	0.497	0.495
range over seeds	0.504–0.548	0.177–0.195	0.744–0.768	0.476–0.509	0.494–0.496
95% interval	[0.482, 0.568]	[0.155, 0.217]	[0.736, 0.774]	[0.444, 0.541]	–
Cycle R^2 of $\mathbf{w} (\approx 0)$	0.006	0.003	0.001	0.000	–0.001
range over seeds	0.003–0.009	0.001–0.006	0.000–0.002	–0.000 to 0.002	–0.004 to 0.002
95% interval	[–0.003, 0.016]	[–0.001, 0.010]	[–0.001, 0.004]	[–0.011, 0.010]	–
I(niche; $\mathbf{w}$) excess $\uparrow$	0.47	0.59	0.60	0.48	–
range over seeds	0.45–0.50	0.58–0.62	0.58–0.62	0.42–0.52	–
95% interval	[0.42, 0.53]	[0.55, 0.66]	[0.55, 0.65]	[0.38, 0.57]	–
Relocation $\uparrow$	0.65	0.72	0.67	0.73 [†]	–
range over seeds	0.63–0.67	0.68–0.75	0.65–0.68	0.71–0.75	–
95% interval	[0.60, 0.69]	[0.65, 0.78]	[0.64, 0.69]	[0.63, 0.85]	–
Atlas cross-seed cosine $\uparrow$	0.97	0.84	0.82	0.91	–
range over seeds	0.97–0.98	0.80–0.90	0.76–0.90	0.89–0.92	–

Marker pairs. After resolVI [10]: twelve marker pairs that should not be co-expressed and six or seven control pairs that are, fixed per section before the read, are scored on held-out cells. A pair is double positive in the raw counts when both genes have a count, and in the corrected decode when both of $(1 - \kappa) \ell_i \rho_{ig}$ round to at least one. The marker-pair contrast, a trajectory (Sec. S7.1), is the change in the exclusive pairs’ double-positive rate from the raw counts to the corrected decode, minus the same change for the control pairs.

S4.2 Merged Subtypes

Within each lineage that absorbed at least two curated classes or clusters, those become sublabels (each with at least 20 training and 10 held-out cells). Probes fitted on training tiles predict the sublabel on held-out tiles from $\boldsymbol{\mu}_z$, $\boldsymbol{\mu}_w$, the prior mean $m_\psi(\mathbf{c}_i, t_i)$ and both latents: a class-balanced ridge classifier (primary) and, per latent, a small perceptron. Two scores are read: per lineage the *balanced accuracy* (the main text’s figure; chance $1/K$ for K sublabels), and per sublabel the one-versus-rest *AUC* against the rest of its lineage, with a floor from five within-lineage permutations of the sublabels and a tile-bootstrap interval, on which the call is made. The call, fixed before the first read: a channel recovers a sublabel when the lower end of its interval is above its floor; channel a is far ahead of b when the interval of their difference is above zero and b ’s excess over its floor is at most half of a ’s; the call is the majority over seeds. Only the curated sublabels of the

primary section and of the TMA core carry an expected allegiance (a state to $\mathbf{z}$, a location to the response); the clusters of the lung and fresh-frozen sections are read for information. The primary section’s cyst lining is its own lineage, so every channel reads it through the type and its allegiance cannot be tested.

Table S13: Merged subtypes on the primary section: balanced accuracy of the held-out ridge classifier of the sublabel within each lineage, by channel. Mean over three seeds, range in brackets. K : number of sublabels; chance is $1/K$. The tumour lineage holds three curated states and the unqualified tumour class.

Lineage (K)	μ_z	μ_w	$m_\psi(\mathbf{c}_i, t_i)$	$\mu_z \oplus \mu_w$	Chance
Fibroblasts (2)	0.66 [0.62, 0.68]	0.95 [0.93, 0.97]	0.94 [0.93, 0.96]	0.95 [0.94, 0.97]	0.50
Endothelial cells (2)	0.67 [0.65, 0.68]	0.90 [0.88, 0.91]	0.89 [0.88, 0.90]	0.90 [0.88, 0.92]	0.50
Tumour (4)	0.79 [0.78, 0.80]	0.41 [0.40, 0.44]	0.39 [0.32, 0.45]	0.81 [0.80, 0.83]	0.25

On the primary section (Table S13) the response separates the tumour- from the stroma-associated fibroblasts and endothelial cells, and its prior mean does so nearly as well; $\mathbf{z}$ reads the tumour states. A perceptron on μ_z reads more of the locations (0.71 and 0.78) and of the states (0.85) without changing the dissociation. By AUC, the locations are read from the response at 0.96 to 0.99 against 0.71 to 0.72 from $\mathbf{z}$, called on two seeds of three, and the proliferating and inflammatory states from $\mathbf{z}$ at 0.96 to 0.97 against 0.61 from the response, on every seed. The exceptions are states that also occupy a region: VEGFA⁺ tumour cells are read from $\mathbf{z}$ (0.96) and also from the response (0.82), and on the TMA core the myofibroblasts among adventitial fibroblasts are read better from the response than from $\mathbf{z}$ (0.92 against 0.77). On the fresh-frozen section three tumour-state clusters are read from both (0.90 to 0.96 from $\mathbf{z}$, 0.77 to 0.86 from the response). The response and its prior mean give nearly the same AUC for every sublabel on the ovarian FFPE, fresh-frozen and TMA sections (within 0.07, 0.01 and 0.06), the per-cell channel being closed (Sec. 3.3); the one TMA exception is activated against venous endothelium, which the prior mean separates better (0.76 against 0.59, on the one gradable seed). On the lung section the response separates clusters within three lineages better than its prior mean (AT1, 0.67 against 0.55; multiciliated cells, 0.71 against 0.57; one tumour cluster, 0.73 against 0.59), although its own deviation adds no more to reconstruction there than elsewhere (Table S23) and $\mathbf{z}$ separates all three better still (0.90): there the response readouts are not only readouts of the context regression.

S5 Simulation

S5.1 Synthetic Recovery

We check, on data simulated from the model’s own generative story, that the final configuration recovers the planted intrinsic states, responses and gene programmes, that the adversary keeps the planted niche out of $\mathbf{z}$, and how much a wrong spill-over fraction costs.

Simulation and fit. $N = 6,000$ cells in a $1,000 \mu\text{m}$ box with $K = 8$ spatially clustered types; the graph, kernel and composition built as for real data, with unit face lengths; $\mathbf{z}_i \sim \mathcal{N}(\mathbf{m}_{t_i}, 0.5^2 \mathbf{I})$; a planted response $\mathbf{w}_i = \mathbf{M}^\top \mathbf{y}_i + 0.3 \boldsymbol{\varepsilon}_i$ linear in the neighbour composition (R^2 0.75 to 0.87); $\log \boldsymbol{\rho}_i = \mathbf{A} \mathbf{z}_i + \mathbf{B} \mathbf{w}_i$ (up to normalisation); the mixture at the planted κ ; $\mathbf{x}_i \sim \text{Multinomial}(\ell_i, \mathbf{p}_i)$ with $\ell_i \sim \max(\text{Poisson}(150), 20)$; and an eight-dimensional image descriptor $\boldsymbol{\Phi}_i = \mathbf{P}^\top \mathbf{y}_i + 0.4 \boldsymbol{\varepsilon}'_i$. $G = 60$ genes, planted $d_z = 4$, $d_w = 2$; three worlds at each planted $\kappa \in \{0, 0.2\}$, with no isolated cells. The fit is the final configuration (Table S4) with widths reduced to the small world and $\alpha_z = \frac{1}{2}/\bar{\ell}$ at the simulated median count, three model seeds per world (nine fits per arm); the uncontrolled arm sets $\alpha_a = 0$, and the misspecified arms fit the worlds planted at $\kappa = 0.2$ at each assumed $\kappa \in \{0, 0.1, 0.2, 0.3, 0.4\}$. The readouts are defined in the caption of Table S14; the loadings’ reference is random two-dimensional subspaces of $\mathbb{R}^{60}$ (cosine mean 0.23, 95th percentile 0.37 to 0.38).

Result. At the matched spill-over fraction every planted quantity is recovered on every world (Table S14). The agreement of μ_z with the planted type is at or above what the counts support (below it in one fit of

Table S14: Recovery on simulated sections at the final configuration, scaled to the simulated world. Three worlds per planted spill-over fraction, each fitted with three model seeds; the range over the nine fits (*Fits*). NMI of the intrinsic latent’s posterior mean with the planted type; *CCA*: the first canonical correlation of the response’s posterior mean with the planted response; *Loadings*: the principal-angle cosine of the fitted against the planted loadings; *Comp. R^2* : the within-type R^2 of neighbour composition from the intrinsic latent. *Misspecified κ* : the worlds planted at $\kappa = 0.2$, fitted at each assumed rate (the matched rate in bold). *References*: NMI of clusters of the counts and of the planted intrinsic state, the composition R^2 of the planted state, and the loading cosine of a random subspace (its mean and 95th percentile), each as the range over the three worlds and both planted rates. A fit is stopped early on 15% of held-out tiles; every read covers all cells. Arrows give the preferred direction.

Arm	κ		Fits	NMI, $\mu_z \uparrow$	CCA, $\mathbf{w} \uparrow$	Loadings $\uparrow$	Comp. $R^2 \downarrow$
	planted	assumed					
<i>Assumed $\kappa = \text{planted } \kappa$</i>							
final configuration	0	0	9	0.80–0.96	0.85–0.95	0.63–0.99	0.00–0.04
without the adversary	0	0	9	0.71–0.94	0.29–0.91	0.14–0.78	0.11–0.19
final configuration	0.2	0.2	9	0.81–0.96	0.83–0.94	0.77–0.99	0.00–0.02
without the adversary	0.2	0.2	9	0.71–0.93	0.38–0.90	0.18–0.85	0.12–0.19
<i>Misspecified κ, planted $\kappa = 0.2$, final configuration</i>							
assumed κ	0.2	0	9	0.86–0.96	0.78–0.94	0.75–0.96	0.00–0.01
assumed κ	0.2	0.1	9	0.82–0.96	0.84–0.94	0.78–0.99	0.00–0.01
assumed κ	0.2	0.2	9	0.81–0.96	0.83–0.94	0.77–0.99	0.00–0.02
assumed κ	0.2	0.3	9	0.80–0.97	0.83–0.94	0.74–1.00	0.00–0.01
assumed κ	0.2	0.4	9	0.79–0.97	0.83–0.94	0.71–1.00	0.00–0.01
<i>References of the simulated worlds</i>							
clusters of the counts			–	0.69–0.93	–	–	–
planted $\mathbf{z}$			–	0.91–0.98	–	–	0.00
random subspace, mean			–	–	–	0.23	–
random subspace, 95th percentile			–	–	–	0.37–0.38	–

eighteen, 0.80 against 0.83) and short of the planted $\mathbf{z}$, which 150 counts per cell cannot reach. The loadings beat the random reference in every fit but vary between seeds on some worlds (one fit at $\kappa = 0$ reaches only 0.63), which is why no gene-level statement on tissue rests on one fit (Sec. 2.4). Without the adversary the intrinsic state carries the niche (composition R^2 0.11 to 0.19 against at most 0.04) and the response and loadings are recovered worse and less reliably. Fitted at any wrong κ from 0 to 0.4 on worlds planted at 0.2, recovery moves little (Fig. S3; canonical correlation lowest at an assumed $\kappa = 0$). Recovery therefore does not point to the planted spill-over fraction, and it says nothing about spill-over of a form the model cannot represent.

S5.2 A Planted Per-Cell Response

At the operating point the response posterior adds about one part in a thousand of held-out reconstruction (Table S23), and its per-cell divergence does not reproduce between seeds (Sec. S6.3). Either these tissues carry little per-cell response beyond what the niche predicts, or the weight $\alpha_w = 0.1$ closes the channel whatever the data; on tissue the two cannot be told apart, so we plant one.

The plant. On the three worlds planted at $\kappa = 0.2$, in 10% of the cells of the most abundant type, drawn at random and so spread across niches, the log-rate is shifted along the first axis of the planted response programme, with norm s times the size of the planted niche effect, for $s \in \{0, 0.5, 1, 2\}$. The subset cannot be predicted from composition and the image descriptor (cross-validated AUC 0.44 to 0.54). The shift lies inside the response programme on purpose: it is a cell responding more or less strongly than its niche predicts, which is what the per-cell channel exists to carry. The fit is the final configuration unchanged, three seeds on world 0 and, as a replication outside the rule, on worlds 1 and 2.

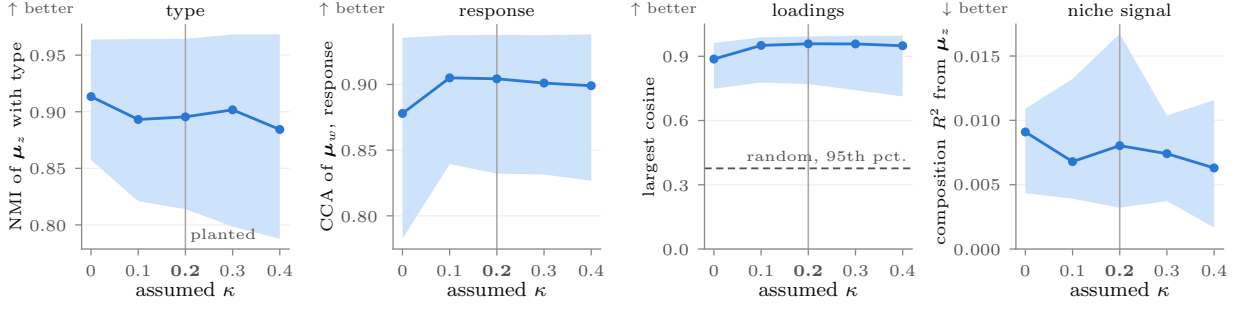


Fig. S3: Recovery on the worlds planted at $\kappa = 0.2$, fitted at each assumed κ (vertical line: the planted value). Panels as the columns of Table S14; dashed: the 95th percentile of random subspaces. Line: mean over nine fits (three worlds, three seeds); band: their range, mostly the spread between worlds. Arrows point toward the better direction.

The rule, fixed in advance. Strength s is detected if, on world 0, every seed separates the planted cells from the rest of their type by the response divergence $\text{KL}(q(\mathbf{w}_i) \| p(\mathbf{w}_i | \mathbf{c}_i, t_i))$ with an AUC of at least 0.8, and every pair of seeds ranks the divergences with a Spearman correlation of at least 0.5. If none were detected, that would be reported: at this weight the channel is closed by design, not by the data.

Result. No strength is detected, on world 0 or on either replication (Table S15). The divergence separates the planted cells only partly and not on every seed (AUC 0.66 to 0.85 at $s = 1$, 0.56 to 0.69 at $s = 2$ on world 0), and a plant does not make seeds agree more (Spearman 0.39 to 0.41 at $s = 0$, lower or similar under a plant; even the null misses the agreement criterion). The response’s deviation from its prior mean moves toward the plant only weakly (correlation 0.11 to 0.41), with a held-out gain below 0.02 nats per count on the planted cells. The plant goes increasingly to the intrinsic state instead: the correlation of μ_z with the plant along the planted direction rises from at most 0.05 at $s = 0$ to 0.36 to 0.48 (world 0), 0.86 to 0.88 (world 1) and 0.52 to 0.70 (world 2) at $s = 2$. At this weight the per-cell channel is therefore closed by design, not by the data: a per-cell response twice the size of the niche effect is not found in the response’s divergence, and much of it goes to $\mathbf{z}$. The absence of reproducible per-cell structure on tissue (Sec. S6.3) thus says nothing about whether the tissue has any; what opening the channel costs is in Sec. S7.4.

S5.3 The Amortisation Gap

The intrinsic encoder $q(\mathbf{z}_i | \mathbf{x}_i, t_i)$ does not read the spill-over influx $\bar{\rho}_i$, so its posterior mean can differ from the exact $\mathbb{E}[\mathbf{z}_i | \mathbf{x}_i, t_i, \bar{\rho}_i]$. This gap cannot be measured on tissue, so we measure it on simulated sections where the exact posterior is computable.

Simulation and fit. DISCELL’s generative model with the response switched off, so a gap is one of inference, not of model mismatch: 3,000 cells in a $700 \mu\text{m}$ square, four types in spatial patches; $\mathbf{z}_i \sim \mathcal{N}(\mathbf{m}_{t_i}, 0.5^2 \mathbf{I})$ in two dimensions and $\rho_i = \text{softmax}(\mathbf{z}_i \mathbf{A})$ over forty genes; the model’s kernel on a Delaunay graph pruned at $40 \mu\text{m}$ ($\tau = 20 \mu\text{m}$, unit faces) at $\kappa = 0.2$; depths $\ell_i \sim \text{Poisson}(D)$, at least 20, for $D \in \{100, 300, 1,000\}$; and an image descriptor made from the neighbour composition plus noise. Given $\bar{\rho}_i$ and the planted parameters, the two-dimensional posterior of $\mathbf{z}_i$ is computed to numerical precision on a 129×129 grid. DISCELL is fitted with its response channel present, at the final configuration reduced to the small world, at the true κ and at κ off by up to 0.1 either way, three seeds each (45 fits). The encoder’s means are mapped onto the simulation’s coordinates by a small cross-fitted network ($\mathbf{z}$ is identified only up to a smooth invertible map), and a cell’s gap is the distance to the exact mean in exact posterior standard deviations.

Result. At the true κ the encoder’s gap is 0.65, 1.02 and 1.93 posterior standard deviations at the three depths (Table S16): above the perceptron trained on the same inputs at every depth (the encoder below it in

Table S15: A planted per-cell response on simulated sections (planted $\kappa = 0.2$, final configuration unchanged). In 10% of the cells of one type, spread across niches, the log-rate is shifted along the planted response programme by s times the size of the planted niche effect (*Shift*, its norm in log-rate units); $s = 0$ is the null. *KL AUC*: the per-cell divergence of the response posterior from its prior separating the planted cells from the other cells of their type; *w corr.*: correlation of the response’s deviation from its prior mean, along the planted direction, with the plant; *z absorption*: the same for the intrinsic latent, read through the decoder along the planted direction; *Recon. gain*: held-out gain in nats per count of the full decode over the response at its prior mean, on planted cells; *Spearman* and *top-decile overlap*: agreement of the per-cell divergence between seeds. Ranges over three model seeds (seed pairs for the last two). The rule, fixed in advance and applied to world 0: the channel detects at s if every seed has KL AUC ≥ 0.8 and every seed pair a Spearman correlation ≥ 0.5 ; worlds 1 and 2 replicate. No strength is detected. The planted subset cannot be predicted from neighbour composition or from composition and the image descriptor (five-fold cross-validated AUC 0.44–0.54 within the planted type, logistic and forest classifiers, all three worlds). The arrow gives the preferred direction.

s	Shift	KL AUC $\uparrow$	w corr.	z absorption	Recon. gain	Spearman	Top-decile overlap	Detected
<i>World 0 (decides)</i>								
0	0.0	0.45–0.51	−0.01 to 0.01	0.00–0.03	0.002–0.006	0.39–0.41	0.42–0.46	no
0.5	4.7	0.38–0.59	0.18–0.28	0.15–0.17	0.005–0.009	0.34–0.36	0.37–0.40	no
1	9.3	0.66–0.85	0.34–0.37	0.23–0.27	0.008–0.017	0.11–0.18	0.21–0.31	no
2	18.6	0.56–0.69	0.16–0.38	0.36–0.48	0.011–0.015	0.25–0.42	0.32–0.41	no
<i>World 1 (replication)</i>								
0	0.0	0.46–0.48	0.02–0.04	−0.00 to 0.04	0.003–0.006	0.41–0.45	0.34–0.38	no
0.5	4.1	0.62–0.78	0.33–0.35	0.27–0.30	0.004–0.013	0.35–0.39	0.34–0.40	no
1	8.1	0.67–0.80	0.32–0.41	0.58–0.63	0.004–0.013	0.33–0.39	0.39–0.42	no
2	16.3	0.46–0.62	0.18–0.21	0.86–0.88	0.001–0.003	0.37–0.39	0.39–0.42	no
<i>World 2 (replication)</i>								
0	0.0	0.46–0.49	−0.01 to 0.05	0.03–0.05	0.000–0.001	0.19–0.25	0.23–0.28	no
0.5	2.7	0.47–0.55	0.12–0.16	0.13–0.18	0.000–0.001	0.19–0.26	0.24–0.31	no
1	5.5	0.56–0.59	0.11–0.29	0.28–0.33	0.002	0.14–0.20	0.23–0.25	no
2	11.0	0.58–0.74	0.20–0.29	0.52–0.70	0.000–0.006	0.16–0.24	0.17–0.33	no

Table S16: Amortisation gap at the true spill-over fraction: distance between an estimate of the posterior mean of $\mathbf{z}$ and the exact one, in exact posterior standard deviations, averaged over cells, mean of three seeds, by depth D . Only the DISCELL encoder is trained without the exact posterior.

Estimate of the posterior mean	Depth D		
	100	300	1,000
DISCELL encoder	0.65	1.02	1.93
Perceptron, the encoder’s inputs	0.61	0.95	1.60
Perceptron, the encoder’s inputs and $\bar{\rho}_i$	0.53	0.80	1.35
Linear regression, the encoder’s inputs	1.39	2.50	4.53
Type mean	2.4	4.1	7.4

only three of nine depth–seed combinations), 0.12 to 0.58 above the perceptron that also sees $\bar{\rho}_i$, and well below the linear regression and the type mean. The gap grows with depth only in units of the narrowing posterior (standard deviation 0.17 to 0.06); in units of $\mathbf{z}$ it stays at 0.16 to 0.17. A wrong κ enlarges it by a factor of at most 1.13, but the gap is not smallest at the true value at any depth (at $D = 1,000$, κ too low by 0.1 gives 1.77 against 1.93). At the development configuration (a closed-form penalty in place of the adversary, 600 fixed epochs, every tile trained) the gap was smaller, 0.55, 0.83 and 1.48, and below the perceptron’s; whether the adversary or the stopping rule accounts for the difference has not been tested. The references have finite capacity and are not information bounds, so these statements are comparative.

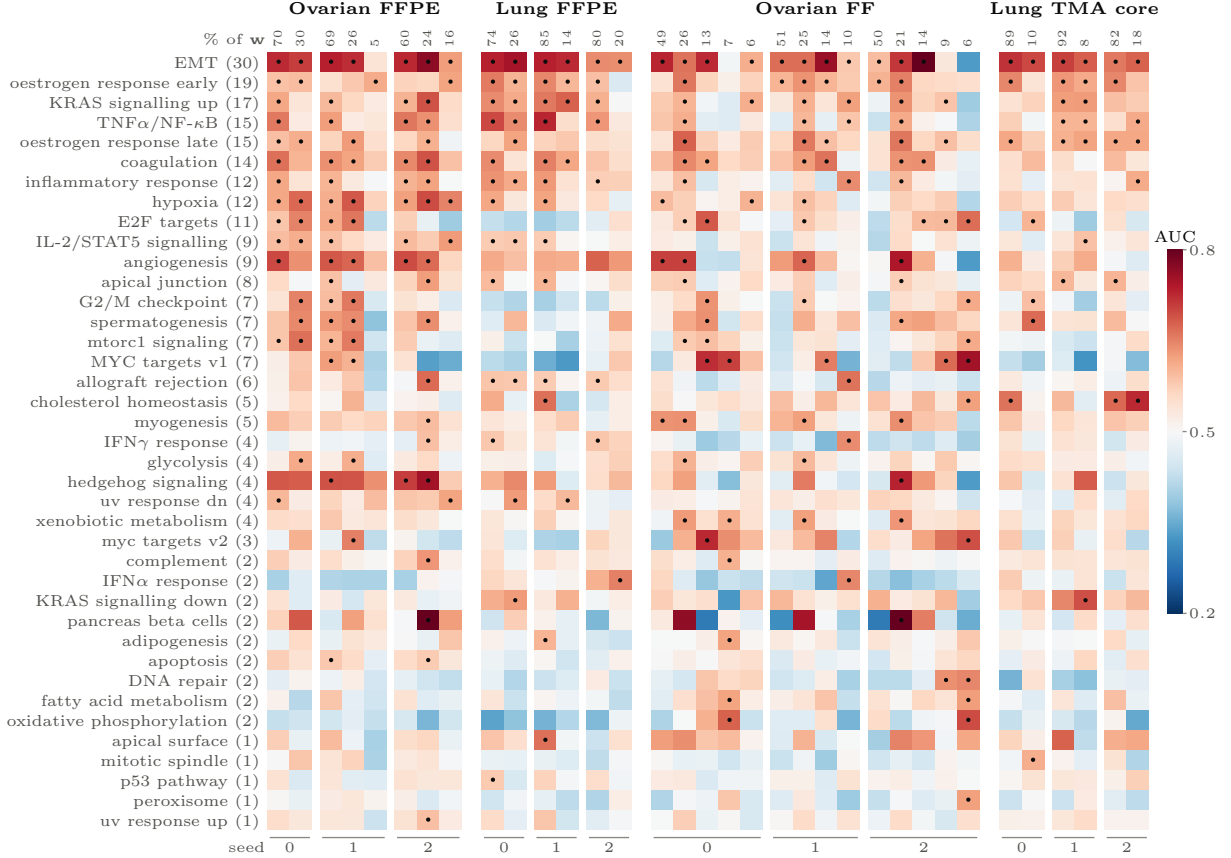


Fig. S4: Hallmark labels of every response programme, by section, seed and programme (within a fit in order of variance; its share of $\mathbf{w}$'s within-type variance, in %, above). Rows: the sets significant on at least one programme (number of such programmes in parentheses). Cells: AUC of the Mann-Whitney test of the set's absolute loadings against the rest of the expressed panel (0.5 is the null); dot, $q \leq 0.05$ (Benjamini-Hochberg within each programme) with AUC above 0.5. Programmes are identified only up to rotation, so columns are compared within a section by the recurrence of their labels, not as matched axes.

S6 Results on Tissue in Full

S6.1 What the Latents Carry and the Response Programmes

Niche information of both latents. With the k -means niches, the response carries 0.41 to 0.63 nats of information about the niche beyond its floor and the intrinsic state 0.06 to 0.11, by the nearest-neighbour estimator, over three seeds of all four sections; of the floor-corrected total, 85 to 90% lies in the response. The neural estimator agrees. These estimates use a different cell sample and floor from Table S12.

Hallmark labels. Fig. S4 shows the enrichment of all 50 hallmark sets on all 34 programmes of the twelve fits; 39 sets are significant on at least one. EMT is significant on 30 of the 34 programmes, followed by oestrogen response early (19), KRAS signalling up (17), $\text{TNF}\alpha/\text{NF-}\kappa\text{B}$ signalling and oestrogen response late (15 each) and coagulation (14); 14 of the 39 sets are significant on at most two programmes.

Localisation. The first programme leans toward extracellular products (rank-biserial effect 0.08 to 0.22) and away from nuclear ones in every seed of the ovarian FFPE, lung and TMA sections; on the fresh-frozen section the extracellular effect is 0.17, 0.06 and 0.00 in the three seeds, fading to null. A gene-specific spill-over

of transcripts near the membrane would produce the same lean, and κ carries no gene index (Sec. 2.1), so the lean does not distinguish a response from spill-over.

Signalling genes. In the spill-over channel, ligand and receptor genes take a larger share of the between-niche shift than other genes on every section and seed (rank-biserial 0.18 to 0.28, $p < 10^{-13}$); in the response channel on the ovarian FFPE, lung and TMA sections (0.12 to 0.18, $p < 10^{-5}$), but not on the fresh-frozen section (0.01 to 0.03, $p = 0.18$ to 0.61). MintFlow’s criterion is therefore met by the spill-over channel alone on every section. The response’s lean breaks down at $\kappa = 0.2$ on the fresh-frozen section (Table 4).

True and false tumour axis. On the primary section the response-predicted shifts follow the tumour bands with a mean $|\tau|$ of 0.78 to 0.81 and the false axis with 0.51 to 0.56. The contrast holds in tumour cells and macrophages and is reversed in the merged fibroblast class (0.63 to 0.71 against 0.71 to 0.75), whose stromal half lies in deep stroma where the false coordinate follows the bands. Shifts predicted from the intrinsic state (0.59 to 0.65 against 0.39 to 0.47), from depth (0.57 to 0.68 against 0.36 to 0.42) and the observed shift (0.47 against 0.33 to 0.36) show contrasts of similar size: the response is closest to the bands, but the band ordering is not specific to it.

Marker pairs. Exclusive double positives are 0.3 to 2.2% of held-out cells on raw counts and control double positives 1.6 to 11.8%. Decoding alone lowers the exclusive rate on every section and raises the control rate on three (lowers it on the primary section), which sets the sign of this trajectory (Sec. S7.1). At $\kappa = 0.1$ the spill-over correction lowers both rates further (exclusive by 0.10 to 0.50 points, control by 0.18 to 1.13) and their ratio on every section, but the data show no removal specific to exclusive pairs, and the rate falls steadily with κ with no plateau, so it does not identify the spill-over fraction.

S6.2 Without the Adversary

Within-type Moran’s I of the posterior means (each coordinate centred on its type’s mean, on the model’s graph restricted to target cells, averaged with weights equal to the coordinates’ within-type variance; permutation null at most 0.005) shows what the adversary removes. Without it ($\alpha_a = 0$, two seeds), I of μ_z is 0.09 to 0.47; with it, 0.05 to 0.35 (Fig. 3d), lower by 42 to 48% on three sections and by 25% on the fresh-frozen one. What remains is within-type variation shared by neighbours that composition and the masked image do not predict (Sec. S2.1). Without the adversary the response’s posterior mean has almost no within-type variance: summed over coordinates, below 10^{-9} in both fresh-frozen fits, 10^{-5} and 0.02 on the ovarian FFPE section and 0.03 to 0.12 on the lung and TMA sections, against 3.0 to 39 in the final fits. The intrinsic state carries the niche instead.

S6.3 The Response at the Type Level

The per-cell divergence of the response from its prior is small, 0.004 to 0.09 nats per cell, and differs up to ten-fold between seeds. Within one fit it is spatially organised (Moran’s I 0.15 to 0.85), but which cells deviate does not reproduce between seeds (per-cell rank correlation -0.43 to 0.57). Under a rule fixed in advance (odds ratio at least 1.5 or standardised difference at least 0.3, interval clear of the null, in all three seeds), the top decile differs from the other cells only coarsely: it avoids tumour on the two FFPE sections and lies among fibroblasts (ovarian) and chondrocytes (TMA core); nothing passes on the fresh-frozen section and no continuous feature passes anywhere. The divergence is therefore not a per-cell anomaly score at the operating point, and a planted per-cell response is not detected at this weight either (Sec. S5.2). The posterior standard deviation of $\mathbf{w}$ is 0.98 to 1.01: the response posterior is its prior in spread as well as in location. The intrinsic state’s divergence and standard deviation do reproduce between seeds (rank correlations 0.78 to 0.93 and 0.53 to 0.93); deeper cells have smaller divergences and wider posteriors (pooled $\rho = -0.77$ to -0.39 and 0.21 to 0.73), the per-cell scaling of Sec. S1.2. The intrinsic state is thus regularised less in low-count cells, and a deep cell’s posterior width overstates its uncertainty, one more reason the reported uncertainty comes from the sweep and the seeds rather than from q .

Table S17: The counterfactual claim head to head: DISCELL’s relocation prediction against Cellina’s neighbour-rewiring counterfactual, scored with the same relocation reads (published read, as in Table S11). Cellina is refitted on the training tiles only, decoded at its posterior mean, and read on the panels, niches, ceilings and bootstrap draws of DISCELL’s first seed, which is shown beside it with the 95% intervals (half-tile subsampling for the fraction, tile bootstrap for Read A; to two decimals); DISCELL’s range over three seeds is in Table S18. *Panels*: panels in the trusted tier. *Read A*: the median gap to the target closed; twin margin: how much closer a cell’s prediction is to its own target than to a random cell’s. *Parts of the prediction*: DISCELL’s prediction from the response programmes alone and from the spill-over alone (the parts are not additive). ^aNot mapped: Cellina has no spill-over channel, so its counterfactual cannot be split into these parts. Arrows give the preferred direction (the parts have none); bold marks the best value per section, of the two estimates. Unassigned cells are not targets.

	Ovarian FFPE		Lung FFPE		Ovarian FF		TMA core	
	DISCELL	Cellina	DISCELL	Cellina	DISCELL	Cellina	DISCELL	Cellina
<i>Fraction of the noise ceiling, trusted tier</i> ↑								
estimate	0.650	0.731	0.743	0.703	0.677	0.566	0.711	0.899
95% int.	[0.62, 0.68]	[0.71, 0.75]	[0.71, 0.78]	[0.67, 0.73]	[0.66, 0.69]	[0.55, 0.58]	[0.63, 0.80]	[0.82, 1.00]
panels	12	12	10	10	169	169	1	1
<i>Read A, own target</i> ↑								
estimate	0.789	0.681	0.826	0.757	0.823	0.602	0.814	0.801
95% int.	[0.76, 0.80]	[0.65, 0.69]	[0.81, 0.84]	[0.74, 0.77]	[0.80, 0.83]	[0.59, 0.62]	[0.74, 0.84]	[0.76, 0.82]
<i>Twin margin, own target</i> ↑								
estimate	0.398	0.361	0.432	0.435	0.437	0.330	0.464	0.284
<i>Parts of the prediction, fraction of the noise ceiling, trusted tier</i>								
programmes	0.520	not mapped ^a	0.535	not mapped ^a	0.535	not mapped ^a	0.641	not mapped ^a
spill-over	0.288	not mapped ^a	0.390	not mapped ^a	0.294	not mapped ^a	0.207	not mapped ^a

S6.4 Relocation in Full

Against Cellina. Cellina’s neighbour-rewiring counterfactual [43] is refitted on the training tiles only, decoded at its posterior mean, and scored by our relocation reads on the panels, niches, ceilings and bootstrap draws of DISCELL’s first seed (Table S17). It has no spill-over channel, so its prediction cannot be split into programme and spill-over parts. The comparison is of one Cellina fit with one DISCELL seed; main Table 3 reads it.

On held-out tiles. The published read is cross-fitted over spatial folds drawn from every tile, so most of its scored cells lie in the model’s training tiles. Table S18 scores the same read on held-out tiles only, with every model quantity still estimated from training tiles and the same panels, ceilings and mask. The rule set before the read was to reconsider the published read if the held-out fraction of the ceiling fell outside its interval on any section; it did, in both directions, and the published read was kept, because an in-sample advantage would lower the held-out read throughout and the fraction of the ceiling does not move that way. The single-cell read (Read A) does, on 11 of 12 fits. The held-out pool is about three quarters of the published one, so fewer panels reach the trusted tier and several all-panel fractions exceed one; on the TMA core no panel is trusted on held-out tiles. On held-out tiles Cellina’s fraction of the ceiling stays above DISCELL’s range on the ovarian FFPE section and falls below it on the lung section.

S6.5 The Latents on Every Section

Fig. S5 extends main Fig. 2 to all four sections. The type of panels c and d is chosen by a rule: among types with at least 3,000 cells, the one whose neighbourhoods are most mixed. UMAP distorts distances, so the figure illustrates what the probe (Table S9) measures and replaces it nowhere.

Table S18: Relocation scored on held-out tiles only, beside the published read. *Published*: the read of Table S11, whose readout is cross-fitted over spatial folds drawn from every tile, so most scored cells lie in the model’s training tiles. *Held-out tiles*: only cells of the model’s held-out tiles are scored, and every model quantity is estimated from its training tiles; same niches, panels, ceilings and Unassigned mask. Fraction of the noise ceiling recovered by the mean read, for panels in the trusted tier and for all panels; Read A, the median gap to the target closed, own target; the twin margin, own target. For DISCELL the range over three seeds; Cellina is its neighbour-rewiring counterfactual, fitted on the training tiles and read on the same panels as the first seed. The held-out pool is about three quarters of the published one, so fewer panels reach the trusted tier. Against the published 95% interval, the held-out read moves in both directions: in the trusted tier it lies above on 3, below on 4 and inside on 2 of the 9 DISCELL fits with both reads; over all panels, above on 8, below on 2 and inside on 1 of 11. Read A is lower on held-out tiles on 11 of 12 fits; the twin margin moves by at most 0.024. Arrows give the preferred direction. –: no panel in the tier, or no value; †over the seeds with a value.

Method	Trusted tier ↑		All panels ↑		Read A ↑		Twin margin ↑	
	published	held-out	published	held-out	published	held-out	published	held-out
<i>Ovarian cancer (FFPE)</i>								
DISCELL	0.631–0.665	0.572–0.633	0.824–0.905	0.923–1.039	0.753–0.789	0.704–0.766	0.388–0.406	0.382–0.398
Cellina	0.731	0.647	1.058	1.331	0.681	0.605	0.361	0.354
<i>Lung cancer (FFPE)</i>								
DISCELL	0.678–0.752	0.595–0.736	0.851–0.943	0.999–1.148	0.737–0.826	0.679–0.768	0.413–0.432	0.397–0.424
Cellina	0.703	0.486	1.040	1.192	0.757	0.718	0.435	0.441
<i>Ovarian cancer (FF)</i>								
DISCELL	0.651–0.677	0.703–0.716	0.647–0.673	0.674–0.688	0.782–0.823	0.777–0.785	0.436–0.455	0.431–0.447
Cellina	0.566	0.558	0.592	0.583	0.602	0.601	0.330	0.329
<i>Lung TMA core</i>								
DISCELL	0.711–0.755†	–	0.943–1.327	1.087–1.226†	0.697–0.814	0.623–0.683	0.447–0.464	0.444–0.457
Cellina	0.899	–	1.327	–	0.801	0.863	0.284	0.271

S6.6 Training Cost

Table S19 gives the cost of a fit. DISCELL’s time per epoch is pure training; its wall-clock time covers the three final fits including evaluations every five epochs and the early stop, whereas the comparison methods’ times are training only, as recorded for the fits compared here. MintFlow’s whole-section fits took 14.8 (lung FFPE) and 15.8 hours (ovarian FFPE) of wall-clock time, against 3.2 to 7.4 minutes for a whole DISCELL fit of the lung section; the other whole-section SIMVI and MintFlow fits could not be run (reasons in Tables S8 and S19). The cost of an epoch is linear in the number of cells, and peak memory is the resident count matrix plus the spill-over influx over the edges of a tile, which the tile size bounds (Sec. S2.5). The sweep costs eighteen fits per section. Sample size is not analysed beyond the range of sizes in Table S6.

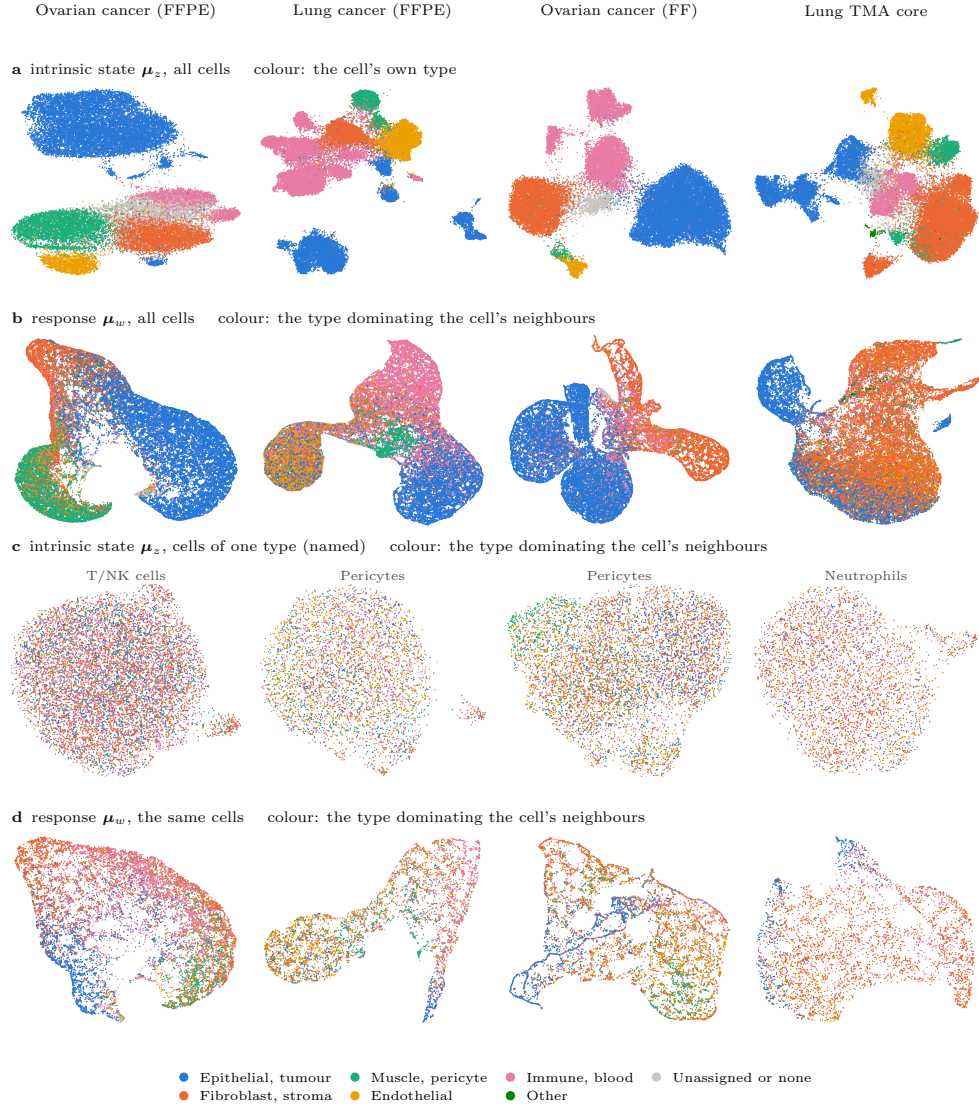


Fig. S5: UMAP of the posterior means, one seed of the final configuration; columns: the four sections. **a**, μ_z of 30,000 random cells, coloured by the cell's lineage family. **b**, μ_w of the same cells, coloured by the family dominating its graph neighbours (grey if none). **c**, μ_z of up to 10,000 cells of one type (named above), coloured as in b. **d**, μ_w of the same cells. Families as in Fig. S2.

Table S19: Model size and training cost on one GPU. *Peak memory*: largest allocation during training, with the section resident on the GPU. *Run time*: one whole fit, for DISCELL including its evaluations every five epochs and the early stop, as the range over the three seeds of the final configuration; for the comparison methods, the training time of the fit compared in this paper, at lineage labels. Every method is fitted on whole sections only. Arrows give the preferred direction; bold marks the best value per section: the fastest finished fit, for DISCELL by its mean. ^aCould not be run on the resources currently available; measured reason: SIMVI, ovarian FFPE: out of GPU memory on a 24 GB card (a further 7.67 GiB requested with 15.87 GiB in use); SIMVI, lung FFPE: out of GPU memory on a 24 GB card (a further 5.16 GiB requested with 21.27 GiB in use); SIMVI, ovarian FF: out of GPU memory on a 24 GB card (a further 40.00 MiB requested with 23.47 GiB in use); MintFlow, ovarian FF: an earlier attempt was stopped for lack of host memory before training (1,157,343 cells, 5,001 genes; an estimated 92 GB needed on a 125 GB host); a retry needs 110 GB free and 102 GB were available.

	Ovarian (FFPE)	Lung (FFPE)	Ovarian (FF)	Lung TMA core
<i>DISCELL, model and training details</i>				
parameters (M)	4.19	4.12	4.11	4.13
peak memory (GiB)	6.3	4.3	15.1	1.4
epochs to the stop	105–115	80–100	175–230	85–90
s per epoch, training only ↓	1.98	1.35	5.78	0.45
<i>Run time of one fit (min) ↓</i>				
DISCELL	8.7–9.3	3.2–7.4	31.2–43.4	1.3
resolVI	32.8	22.0	105.0	5.5
Cellina	11.6	9.9	29.1	3.4
SIMVI	not run ^a	not run ^a	not run ^a	192.6
MintFlow, 50 epochs	938.6	879.4	not run ^a	544.0

S7 Spill-over Sweep and Sensitivity

S7.1 The Claimed Contrasts and the Rules

Table S20 defines the contrasts whose breakdown points the main text reports (Table 4), and Fig. S6 draws them across the grid. Two contrasts first in the graded set were moved to trajectories after reading (Table S21): the marker-pair contrast, which measures the decode rather than the spill-over correction, being already present at $\kappa = 0$, and the relocation’s margin over its programme part, which is zero at $\kappa = 0$ by construction. **[PENDING RUNS: breakdown gaps queue: the families become seven contrasts on each whole-slide section (the tumour axis added on lung and FF), six on the TMA core and six on the serial section, read through the core’s fits; update this paragraph and the serial-section rule below]**

Rules. No readout was designated primary in advance: the final sweeps were preceded by development sweeps, so any such set would have been chosen with knowledge of earlier results. The interval of a contrast is for its mean over seeds, from a spatial block bootstrap over tiles within each seed’s fit, two-sided and Bonferroni-adjusted over the claimed contrasts of the section; rankings are read through pairwise differences, each with its own breakdown point. Findings are section-level. A tissue-specific finding is claimed only if it holds on both serial sections of a tissue where two exist; only the TMA core has one, and only the cycle asymmetry is read on it, so the rule applies to that contrast alone. A property of the method, such as whether a readout moves with κ at all, is required on every section. A decline along the grid is mechanical, since a larger κ assigns more of the neighbours’ resemblance to the mixture; a change in the direction of a programme or in which latent a target aligns with is not. Programmes are identified only up to rotation and per-column constants (Sec. S1.4), so raw loadings and probe directions are never compared across runs. The interval at a grid point is a sampling statement that narrows with more cells; the trajectory across the grid is not, because the confound it expresses is not a sampling problem.

Table S20: The claimed contrasts, named as in Table 4. Each is signed so that a finding is positive; its breakdown point follows Definition 1. The relocation reads are defined in Sec. S4.1, the signalling share and the tumour axis in Sec. S4.1 and Sec. S6.1. The tumour axis needs tumour-proximity bands and is read on the ovarian FFPE section; the serial section reads the cycle asymmetry only, through the core’s fits.

Contrast	Definition	Positive means	Main text
<i>Biological claims</i>			
Relocation > spill-over part	mean R^2 over all panels of the prediction with programmes and spill-over, minus that of the spill-over channel alone	the programmes add to what spill-over predicts	Sec. 3.6
Relocation > type mean	median over panels of the gap closed by the relocated cells (single-cell read, Read A), minus that closed by copies of the target’s mean profile	moved cells match the target’s distribution, not only its mean	Sec. 3.6
Nearest-twin advantage	median over panels of $(d_{\text{random}} - d_{\text{twin}})/d_{\text{random}}$, Hellinger distances from a target cell to its nearest relocated cell in μ_z and to a random one (twin margin)	the intrinsic state picks the right counterpart	Sec. 3.6
Signalling-gene lean of $\mathbf{w}$	rank-biserial effect of the per-gene response share, ligand and receptor genes against the other genes	signalling genes take more of the response	Sec. 3.7
Tumour axis in $\mathbf{w}$	mean $ \tau $ of the response-predicted shift with the tumour-proximity bands, minus that with a false spatial axis	the response follows the bands	Sec. 3.7
<i>Allocation checks</i>			
Cycle in $\mathbf{z}$, not in $\mathbf{w}$	cycle R^2 of μ_z minus that of μ_w , top-decile cycling held-out cells, centred per type	the cycle is in the intrinsic state	Sec. 3.3
Niche information in $\mathbf{w}$	$I(\text{niche}; \mu_w)$ minus its within-type permutation floor, held-out cells, nats	the response carries the niche beyond the type	Sec. 3.3

Every contrast holds across the grid except the signalling-gene lean on the fresh-frozen section, which breaks down at $\kappa = 0.2$, and the relocation’s margin over its spill-over part, which breaks down at $\kappa = 0.4$ on the primary section and 0.3 on the lung section, where a seed changes sign.

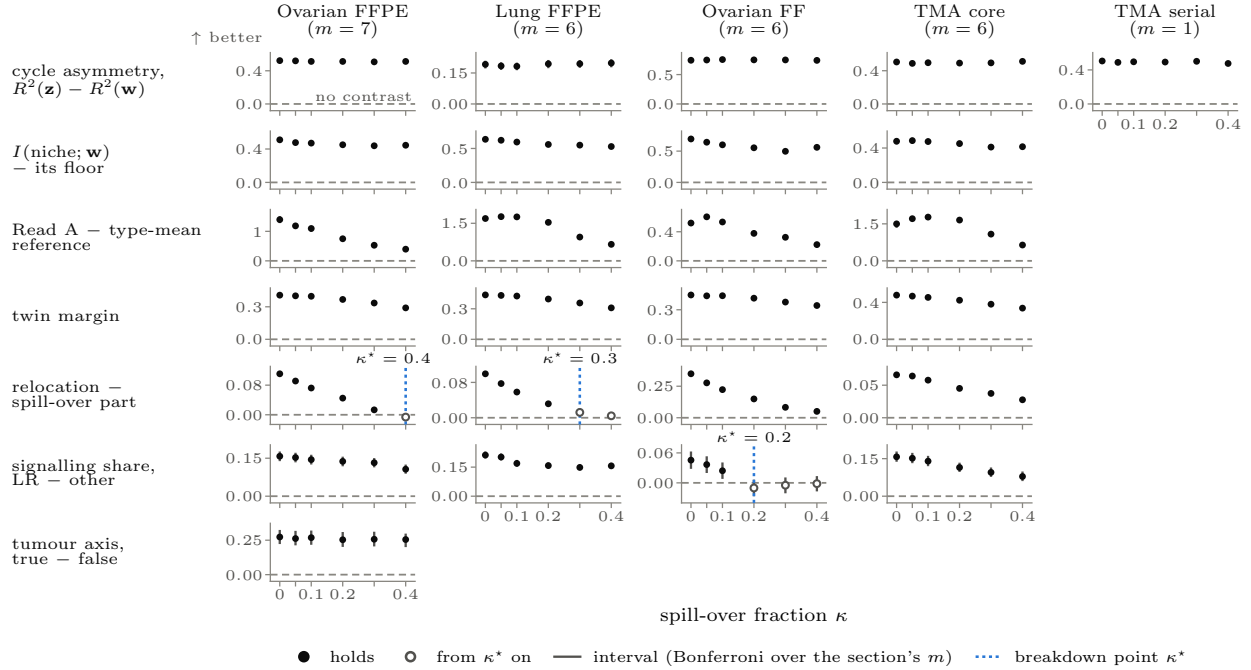


Fig. S6: The claimed contrasts (Table S20) across the spill-over sweep; m , the number of contrasts in the section's set. At each κ , the mean over three seeds and its Bonferroni-adjusted interval; filled: the contrast holds; hollow: from κ^* on (dotted). Each panel has its own vertical scale, which includes zero (dashed); empty panel: not in that section's set.

S7.2 Diagnostics and Trajectories across the Sweep

Fig. S7 draws the diagnostics of every fit of the sweep, and Table S21 the readouts reported as trajectories: magnitudes, positive under any fit, and the two reclassified contrasts. The relocation's margin over its programme part is positive in every seed on every section from $\kappa = 0.05$ to 0.2; it stays positive up to $\kappa = 0.4$ on the lung and fresh-frozen sections and turns negative on the primary section from $\kappa = 0.3$ and, in the mean, on the TMA core at 0.4. The relocation read's fraction of the ceiling is lower at $\kappa = 0$ than at the operating point on every section, peaks at $\kappa = 0.05$ to 0.2 and falls beyond, most on the primary section, where it reaches 0.35 at $\kappa = 0.4$. As κ grows, Moran's I of the intrinsic state falls on every section, while that of the response changes by at most 0.05. The within-type share of the response's variance (0.26 to 0.31 on the primary section, 0.40 to 0.44 on the lung section, 0.53 to 0.56 on the fresh-frozen section, 0.49 to 0.55 on the TMA core) and the effective number of programmes (at most one axis apart on a section) change little, and the overlap of the programme spaces with the reference fit stays at 0.76 to 0.92 on average, comparable to the overlap between seeds at the operating point (lowest member 0.72 to 0.84), with a mild decline at $\kappa = 0.4$.

The held-out probe is graded at every grid point. The nonlinear probe's composition residual, as a seed mean, is 4.0 to 4.7% on every section from $\kappa = 0$ to 0.1 and 2.9 to 4.5% beyond, with single seeds between 2.0 and 6.1% over the whole grid (Table S21). The invariance therefore does not weaken along the grid, and a readout's change with κ is not a change in the residual niche signal.

S7.3 A Planted Spill-over Control

In simulation, a spatial contrast created only by spill-over at a known fraction is planted beside a genuine response contrast, and the sweep is run with the same rule (Sec. 3.7). The prediction, set before the run, is that the spill-over contrast breaks at the first grid point at or above the planted fraction while the response contrast holds; if it does not, the breakdown point is reported as a check of lower power. **[PENDING RUNS: planted spill-over control: design details and result]**

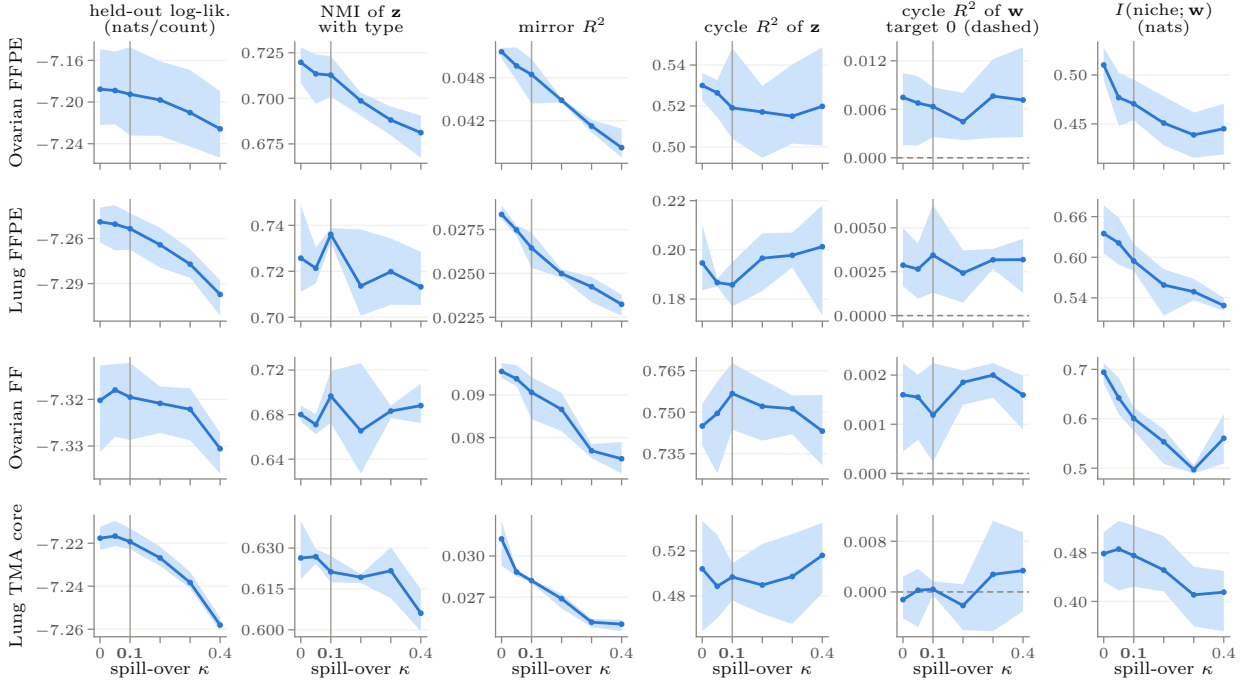


Fig. S7: The diagnostics across the spill-over sweep, per section: held-out reconstruction (nats per count), NMI of $\mathbf{z}$ with the type, mirror R^2 , cycle R^2 of $\mathbf{z}$ and of $\mathbf{w}$ (expected near zero, dashed) on the top-decile cycling set, and $I(\text{niche}; \mathbf{w})$ above its within-type floor (nats). Line: mean over three seeds; band: their range; vertical line: the operating point $\kappa = 0.1$. These readouts move with κ by construction, and κ is not selected on them: $\kappa = 0.1$ matches the typical per-cell misassignment [16].

S7.4 Sensitivity to the Assumptions

Table S22 refits the final configuration with one setting changed at a time, on the primary section and, for two of the families, on the TMA core and the fresh-frozen section, and gives each arm’s move in units of the final configuration’s standard deviation over its three seeds. An arm has two or three seeds, and that standard deviation is itself a rough estimate, so moves of one or two units are within what a change of seed can produce. Nothing is adopted from these arms.

The form of the spill-over. The rate is set per cell, in proportion to the ratio of its neighbours’ depth, or density, to its own (capped), or per gene, in proportion to the gene’s share of transcripts detected outside nuclei, each at the same mean rate as the section-wide κ . No read moves by more than 1.3 standard deviations, and the relocation read stays at 0.65 to 0.66. Forms outside these three are not tested.

A false-positive floor. The model has no background term (Sec. S8). A fixed floor added to the mixture, one rate per section or one per cell scaled by its area, leaves every diagnostic of the primary section within 0.6 standard deviations, but lowers its relocation read from 0.65 to 0.50 (−8.3) with the per-section floor and to 0.54 (−6.2) with the area-scaled one, the largest moves in the table. On the TMA core the floor leaves the relocation read at 0.73 and moves the agreement with type up by 2.4 and the mirror down by 4.0 standard deviations of a very narrow spread. The relocation read on the primary section therefore depends on the absence of a background term, and it is the read to treat with most caution there.

The response weight. The operating point $\alpha_w = 0.1$ lies far above the bound-equivalent value $1/\bar{\ell}$ (Sec. S1.2): $\alpha_w = m/\bar{\ell}$ with m of about 14 on the primary section, 28 on the TMA core and 143 on the fresh-frozen section. Lowering it opens the per-cell channel: at $\alpha_w = 1/\bar{\ell}$ the response’s deviation from its

Table S21: Readouts reported as trajectories across the sweep, mean over three seeds. *Relocation*: fraction of the noise ceiling, trusted tier ([†]over the seeds with a trusted panel). *Marker pairs*: change in the double-positive rate of exclusive pairs from raw counts to the corrected decode, minus that of control pairs, in percentage points (Sec. S4.1). *Relocation – programme part*: mean R^2 over all panels with the spill-over term minus that of the programme part alone. *Probe*: the nonlinear probe’s composition residual, $1 - e^{-2 \text{excess}}$ in % (as in Table S9).

Readout	$\kappa = 0$	0.05	0.1	0.2	0.3	0.4
<i>Ovarian cancer (FFPE)</i>						
Relocation, fraction of ceiling	0.56	0.63	0.65	0.62	0.47	0.35
Relocation – programme part	0.00	0.02	0.02	0.02	−0.01	−0.04
Marker pairs (pp)	0.4	0.4	0.6	0.6	0.7	0.9
Moran’s I of μ_z	0.118	0.112	0.108	0.096	0.083	0.072
Probe, composition (%)	4.7	4.5	4.6	4.5	3.5	3.5
<i>Lung cancer (FFPE)</i>						
Relocation, fraction of ceiling	0.58	0.69	0.72	0.74	0.66	0.61
Relocation – programme part	0.00	0.02	0.04	0.04	0.03	0.02
Marker pairs (pp)	−1.7	−1.6	−1.6	−1.3	−0.8	−0.6
Moran’s I of μ_z	0.064	0.059	0.054	0.048	0.043	0.038
Probe, composition (%)	4.4	4.2	4.4	4.0	3.7	3.7
<i>Ovarian cancer (FF)</i>						
Relocation, fraction of ceiling	0.58	0.64	0.67	0.70	0.67	0.62
Relocation – programme part	0.00	0.06	0.10	0.11	0.12	0.10
Marker pairs (pp)	−2.9	−2.8	−2.5	−2.0	−1.4	−0.9
Moran’s I of μ_z	0.360	0.358	0.351	0.341	0.312	0.302
Probe, composition (%)	4.5	4.3	4.6	4.3	3.5	3.5
<i>Lung TMA core</i>						
Relocation, fraction of ceiling [†]	0.58	0.73	0.73	0.73	0.70	0.67
Relocation – programme part	0.00	0.02	0.02	0.02	0.01	−0.00
Marker pairs (pp)	−1.7	−1.7	−1.5	−1.2	−0.8	−0.6
Moran’s I of μ_z	0.076	0.066	0.059	0.052	0.040	0.038
Probe, composition (%)	4.0	4.1	4.0	3.4	3.4	2.9

prior mean carries 3.5% of its variance on the primary section and 21% on the TMA core, against 0.2% at the operating point, and adds 0.015 to 0.017 nats per count to held-out reconstruction, against at most 0.001. The cost is in the agreement of $\mathbf{z}$ with type, which falls by 2.6 and 2.8 standard deviations (0.684 against 0.713, 0.607 against 0.621), and by 2.9 on the fresh-frozen section at $10/\ell$ (0.629 against 0.697). The lower weights raise the relocation read on the primary section (0.71 at $1/\ell$) and on the fresh-frozen section (0.75 at $5/\ell$ and $10/\ell$), but not on the TMA core. The operating point thus keeps the agreement with type at the price of a closed per-cell channel (Sec. S5.2). The deviation’s share of variance and its held-out gain come from the same fits but are not in the table.

The adversary. Doubling the heads’ update steps or their width, or averaging an ensemble of three heads, changes the nonlinear composition residual by at most 0.7 standard deviations; doubling the steps costs 4.3 in agreement with type. Weighting the composition term equally with the image term instead of three times raises the residual from 4.6% to 6.6% (+2.5) and the mirror by 2.0; a weight of five lowers the residual to 3.7% at a cost of 2.1 in agreement with type.

Without the intrinsic path. The final configuration is refitted on all four sections with $\omega = 0$, three seeds at the final fits’ splits. The prediction, set before the fits: the allocation shifts toward $\mathbf{w}$, so the agreement of $\mathbf{z}$ with type and its cycle R^2 fall, those of μ_w rise, and the within-type share of $\mathbf{w}$ ’s variance changes. A read counts as moved when the mean over seeds lies outside the final fits’ range. Two reads move on every section with disjoint ranges: the NMI of μ_w with the type rises (0.42 to 0.50 on the primary section, 0.27 to 0.32 on the lung section, 0.33 to 0.35 on the fresh-frozen section, 0.29 to 0.40 on the TMA core), and the within-type share of $\mathbf{w}$ ’s variance falls (0.32 to 0.19, 0.43 to 0.37, 0.53 to 0.49, 0.53 to 0.33). The intrinsic state loses less: its cycle R^2 falls on the fresh-frozen section (0.76 to 0.70, disjoint ranges); on the primary section its

Table S22: Sensitivity of the final configuration to the assumptions of the model, each arm refitted with one setting changed. For the final configuration, the mean over its seeds with the range; for an arm, its mean and, in brackets, its move from the final configuration in units of the final configuration’s standard deviation over seeds; *Seeds* gives the number of seeds fitted. Reconstruction is held-out, in nats per count; $I(\text{niche}; \mathbf{w})$ is the excess over the within-type floor, in nats. *MLP*: the residual neighbour composition in the intrinsic latent, as the share of the block’s within-type variance that the nonlinear probe explains beyond the permutation floor, $1 - e^{-2 \text{excess}}$ (as in Table S9), in %. *Relocation*: the fraction of the noise ceiling recovered by the mean relocation read, trusted tier (as in Table S11); [†]over the seeds with a trusted panel. The spill-over fraction is set per cell from its neighbours’ depth or density relative to its own, or per gene, at the same mean value; the false-positive floor is fixed once per section, or per cell scaled by its area. Arrows give the preferred direction, so a move in the direction of a column’s arrow is an improvement. Unassigned cells are not targets.

Arm	Seeds	Recon. $\uparrow$	NMI $\uparrow$	Mirror $R^2 \downarrow$	$I(\text{niche}; \mathbf{w}) \uparrow$	MLP (%) $\downarrow$	Relocation $\uparrow$
<i>Ovarian cancer (FFPE)</i>							
final configuration	3	−7.193	0.713	0.048	0.47	4.6	0.65
range		−7.232 to −7.148	0.701–0.723	0.044–0.051	0.45–0.50	4.0–5.5	0.63–0.67
<i>Form of the spill-over fraction</i>							
per cell, depth ratio	3	−7.202 (−0.2)	0.699 (−1.3)	0.052 (1.1)	0.48 (0.3)	5.3 (1.0)	0.66 (0.6)
per gene	3	−7.193 (−0.0)	0.716 (0.3)	0.048 (−0.3)	0.48 (0.2)	4.4 (−0.2)	0.65 (0.2)
per cell, density ratio	3	−7.192 (0.0)	0.700 (−1.1)	0.047 (−0.3)	0.47 (0.2)	4.4 (−0.3)	0.65 (0.1)
<i>Fixed false-positive floor</i>							
one per section	3	−7.203 (−0.2)	0.718 (0.5)	0.051 (0.6)	0.48 (0.5)	4.8 (0.2)	0.50 (−8.3)
area-scaled, per cell	3	−7.200 (−0.2)	0.712 (−0.1)	0.049 (0.3)	0.47 (−0.2)	4.3 (−0.4)	0.54 (−6.2)
<i>Weight of the response deviation</i>							
$\alpha_w = 1/\bar{\ell}$	2	−7.170 (0.5)	0.684 (−2.6)	0.042 (−1.9)	0.41 (−2.7)	3.4 (−1.5)	0.71 (3.6)
$\alpha_w = 2/\bar{\ell}$	2	−7.186 (0.1)	0.689 (−2.1)	0.046 (−0.7)	0.47 (0.0)	4.1 (−0.7)	0.69 (2.6)
$\alpha_w = 5/\bar{\ell}$	2	−7.186 (0.2)	0.705 (−0.7)	0.046 (−0.6)	0.49 (1.0)	4.2 (−0.5)	0.69 (2.2)
$\alpha_w = 10/\bar{\ell}$	2	−7.190 (0.1)	0.708 (−0.4)	0.046 (−0.6)	0.47 (−0.0)	4.3 (−0.4)	0.64 (−0.2)
$\alpha_w = 25/\bar{\ell}$	2	−7.190 (0.1)	0.716 (0.3)	0.048 (−0.0)	0.44 (−1.5)	5.5 (1.2)	0.62 (−1.6)
<i>Adversary capacity and weight</i>							
head steps 12	2	−7.190 (0.1)	0.665 (−4.3)	0.052 (1.0)	0.45 (−0.7)	4.3 (−0.4)	0.70 (3.1)
head width 128	2	−7.195 (−0.1)	0.701 (−1.0)	0.051 (0.8)	0.46 (−0.6)	5.1 (0.7)	0.67 (1.1)
ensemble of 3	2	−7.191 (0.0)	0.726 (1.1)	0.046 (−0.8)	0.47 (0.0)	4.5 (−0.1)	0.64 (−0.7)
composition weight 1	2	−7.195 (−0.0)	0.710 (−0.3)	0.056 (2.0)	0.44 (−1.5)	6.6 (2.5)	0.62 (−1.5)
composition weight 5	2	−7.187 (0.1)	0.690 (−2.1)	0.047 (−0.5)	0.48 (0.4)	3.7 (−1.1)	0.68 (1.6)
<i>Lung TMA core</i>							
final configuration	3	−7.219	0.621	0.028	0.48	4.0	0.73 [†]
range		−7.223 to −7.213	0.617–0.627	0.028–0.028	0.42–0.52	3.5–4.8	0.71–0.75
<i>Fixed false-positive floor</i>							
one per section	3	−7.218 (0.3)	0.634 (2.4)	0.028 (−4.0)	0.48 (0.1)	4.0 (−0.1)	0.73 (−0.1) [†]
<i>Weight of the response deviation</i>							
$\alpha_w = 1/\bar{\ell}$	2	−7.202 (3.1)	0.607 (−2.8)	0.027 (−5.1)	0.42 (−1.0)	3.5 (−0.7)	0.72 (−0.4) [†]
$\alpha_w = 2/\bar{\ell}$	2	−7.210 (1.8)	0.616 (−1.1)	0.028 (−0.2)	0.46 (−0.3)	3.1 (−1.4)	0.72 (−0.5) [†]
$\alpha_w = 5/\bar{\ell}$	2	−7.215 (0.8)	0.615 (−1.3)	0.028 (−4.3)	0.51 (0.7)	3.7 (−0.5)	0.72 (−0.5) [†]
$\alpha_w = 10/\bar{\ell}$	2	−7.215 (0.7)	0.627 (1.1)	0.028 (−1.0)	0.48 (0.1)	3.2 (−1.2)	0.72 (−0.5) [†]
$\alpha_w = 25/\bar{\ell}$	2	−7.217 (0.4)	0.624 (0.6)	0.029 (5.0)	0.48 (0.2)	3.5 (−0.9)	0.69 (−1.3) [†]
<i>Ovarian cancer (FF)</i>							
final configuration	3	−7.320	0.697	0.091	0.60	4.6	0.67
range		−7.329 to −7.312	0.673–0.719	0.084–0.094	0.58–0.62	4.4–5.0	0.65–0.68
<i>Weight of the response deviation</i>							
$\alpha_w = 5/\bar{\ell}$	2	−7.310 (1.2)	0.660 (−1.6)	0.083 (−1.5)	0.68 (3.3)	4.4 (−0.9)	0.75 (5.3)
$\alpha_w = 10/\bar{\ell}$	2	−7.313 (0.7)	0.629 (−2.9)	0.087 (−0.6)	0.67 (2.8)	4.2 (−1.3)	0.75 (5.3)

agreement with type (0.71 to 0.67) and cycle R^2 (0.52 to 0.50) fall by the mean only; elsewhere both stay within range. The cycle R^2 of $\mathbf{w}$ rises nowhere, no read without a prediction moves with disjoint ranges, and no move is opposite to the prediction. The path’s role is thus to keep the type out of $\mathbf{w}$. One final fit on the TMA core, refitted with the software used for the ablation, comes out close to but not identical to the original (NMI of $\mathbf{z}$ 0.616 against 0.621, just below the final fits’ range); the moves above are larger.

S7.5 What Each Part of the Model Buys in Reconstruction

Held-out reconstruction reads a cell’s counts twice, in the encoder and in the likelihood, so one number of it says little. Table S23 decodes the held-out cells in four ways: the *type profile* (the type’s empirical count profile); the *context* decode, which uses none of the cell’s counts ($\mathbf{z}$ at its type’s mean, $\mathbf{w}$ at the prior mean $m_\psi(\mathbf{c}_i, t_i)$); the *intrinsic* decode (own $\mathbf{z}$, $\mathbf{w}$ at the prior mean); and the *full* decode. The spill-over influx and κ are the fit’s in every decode. The cell’s own intrinsic state carries most of what the model adds over the type profile; the context adds a little on three sections and nothing on the lung FFPE section; the response’s own posterior adds one part in a thousand or less beyond its prior mean, by the operating point’s design (Sec. S1.2), with the lung section the exception on subtype recovery (Sec. S4.2). The model fitted to the TMA core reads its serial section as it reads the core.

Table S23: Held-out reconstruction by what the decode may use, in nats per count: mean over three seeds, with the range. *Full*: log-likelihood per count of the full decode, mean only. The other columns are differences: context over type profile, intrinsic over context (the cell’s own $\mathbf{z}$), full over intrinsic (the cell’s own $\mathbf{w}$). Every difference has a 200 μm tile-bootstrap 95% interval; all exclude zero except context over type profile on the lung FFPE section, and on the TMA core for one seed. Target cells of held-out tiles; for the serial section, every cell, read through the model fitted to the core.

Section	Full	Context	Own $\mathbf{z}$	Own $\mathbf{w}$
Ovarian cancer (FFPE)	−7.19	0.025 (0.019–0.029)	0.133 (0.116–0.154)	0.0008 (0.0008–0.0010)
Lung cancer (FFPE)	−7.25	−0.000 (−0.001 to 0.000)	0.167 (0.163–0.171)	0.0004 (0.0003–0.0005)
Ovarian cancer (FF)	−7.32	0.033 (0.032–0.034)	0.059 (0.057–0.061)	0.0001 (0.0001–0.0001)
Lung TMA core	−7.22	0.007 (0.003–0.013)	0.127 (0.124–0.129)	0.0004 (0.0003–0.0005)
Lung TMA, serial section	−7.24	0.010 (0.008–0.012)	0.127 (0.125–0.129)	0.0004 (0.0003–0.0005)

S8 Limitations

The conclusion names the main limitations; this section collects them all, each with where it is argued.

- *The form of the spill-over.* One fraction per section, the same for every gene, scaled by the receiving cell’s own rate, and no ambient term (Sec. 2.1). A low-count cell beside high-count donors is therefore under-corrected at every κ and the reverse over-corrected; the niche-predictable part of that error lands in $\mathbf{w}$, and the sweep does not address it. Ambient RNA captured by the panel’s own probes is not registered by the negative controls [44] and is not modelled. If extranuclear transcripts, enriched for secreted and membrane proteins [45], cross boundaries more readily, that gene-specific spill-over is spanned by no κ and lands in $\mathbf{w}$, so a programme enriched for cell-surface genes cannot be separated from it by the sweep. Three other forms move no read on the primary section by more than 1.3 seed standard deviations, but a fixed false-positive floor lowers the relocation read from 0.65 to 0.50 (Sec. S7.4).
- *What the sweep separates.* Spill-over of the modelled form from a spatial association, not spatially organised intrinsic variation within a type, which the conditional invariance assigns to the response, nor cells selecting their niche (Sec. 2.4). The bound of Proposition 1 holds with the neighbours’ clean profiles fixed and is not computed on any section, so the grid supports only survival up to $\kappa = 0.4$.
- *Labels.* The labels come from the same counts, spill-over included, and the decomposition is relative to their granularity: a spatially clustered state merged into its lineage, such as a subclone, is read as a response (Sec. S2.1). Label errors caused by spill-over concentrate at heterotypic interfaces and in low-count cells, and enter through t_i and the neighbours’ compositions, where neither κ nor the invariance reaches them.
- *The invariance is incomplete.* A nonlinear probe still recovers 3.5 to 5.5% of the within-type variance of neighbour composition from $\mathbf{z}$ on every section, and on the TMA core the adversary removes only 29 to 48% of what the model without it leaves (Table S9). The development threshold, a quarter of the uncontrolled excess, is not met on composition on three sections and on the image block on the TMA core, so the residual is reported descriptively. A probe at its floor bounds only the dependence it can detect.

- *The response at the operating point.* With α_w far above its bound-equivalent value, the response readouts are in effect readouts of the context regression; the lung section is a partial exception (Sec. S4.2). The per-cell channel is closed by design, not by the data: a planted per-cell response is not detected and is partly absorbed by $\mathbf{z}$ (Sec. S5.2), and lowering α_w opens the channel at a cost in agreement with type (Sec. S7.4). Several response readouts are not specific to it: the ligand–receptor criterion is met by the spill-over channel alone, the lean toward membrane and secreted genes is what gene-specific spill-over would produce, and the tumour band ordering is also followed by $\mathbf{z}$ and by depth (Sec. S6.1).
- *The cycle reference.* The cycle scores are gene-set scores of the cell’s own segmented counts, into which spilled-over transcripts and environment-induced expression enter; the asymmetry between the latents shows where the model places this axis, not that it is the cell’s own (Sec. 3.3).
- *Inference.* The intrinsic encoder does not see the spill-over influx; in simulation its posterior mean is 0.65 to 1.93 posterior standard deviations from the exact one, more at every depth than a perceptron fitted on the same inputs, and on tissue the gap cannot be measured (Sec. S5.3). The objective is a weighted surrogate, so the uncertainty reported comes from the sweep and the seeds, not from the posterior (Sec. S1.2); it is not unimodal in the seed, and fits with a dead response channel were discarded and re-seeded (none among the final fits).
- *Identification.* Beyond the upper bound on κ and the invariance the probe grades, nothing guarantees that the intrinsic and response coordinates are identified (Sec. 2.4).
- *Evaluation.* Cells of held-out tiles enter training as stop-gradient neighbours (Sec. S2.5). Every fit is to one section of one platform, only one section is held out whole, and sample size is not analysed. Most cells scored by the relocation reads lie in training tiles; on held-out tiles the fraction of the ceiling moves both ways and the single-cell read falls (Table S18). The comparison methods ran at their published defaults on our held-out cells, while DISCELL’s weights were calibrated on these sections (Sec. 3.1); two could not be run on some sections (Table S8). Edge features in the attention were not tested.

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